

Stable modal subspaces preserve inlet-related deformation across human pelvic anatomical variability

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Short title: Stable modal subspaces in the human pelvis

Abstract

Human pelvis kinematics must be maintained across marked variation in ring geometry, inlet and outlet dimensions, as well as bone architecture. Yet, population biomechanics commonly compares ranked deformation pathways as if the same spectral label carried the same biological meaning in every individual. Here, the human pelvis has been utilized to assess whether key mechanical functions remain attached to individually stable pathways or are preserved only within small recurrent groups of interchangeable pathways. We analyse the linearised stiffness spectrum of $N=278$ subject-specific finite-element models assembled on a shared reference domain with individual bony geometry and CT-derived bone elasticity. Across the population cohort, pelvic mechanics can be separated into two distinct regimes. Modes 1–3 form a gap-protected low-rank *backbone*, which rarely reorders across subjects ($< 0.4\%$ swap rate) and dominates standing loads. Higher ranks contain recurrent *clustered* regions, most prominently an inlet-related 9–10 block present in 54.7% of individuals. Within that block, anterior–posterior inlet coupling shifts between rank labels 9 and 10 when the labels exchange, yet remains substantial in their shared two-dimensional subspace and is strongly recruited under parturition-motivated proxy loading. An outlet-related 5–6 block and a mixed 12–15 block show that clustered behaviour can preserve either a distinct anatomical program or a broader higher-order reserve. These results indicate that pelvic mechanical function is organised across spectral scales: some features remain tied to stable low-rank pathways, whereas others are preserved only within the smallest recurrent function-preserving subspace. More generally, the analyses offered an approach to identify how biological mechanical function is maintained in anatomically variable structures without assuming one-to-one preservation of spectral labels.

Author summary

Individuals differ substantially in pelvic shape and bone structure, yet the human pelvis still has to perform the same basic mechanical tasks. We asked whether those tasks are carried by one fixed deformation pattern for each individual, or by a small set of similar patterns which can substitute for one another. Using finite-element models of 278 pelvises, we compared how deformation patterns were organised across a population subset. It was found that everyday load transfer relies on a small low-rank backbone shared across subjects. In contrast, some higher-order functions, including opening of the pelvic inlet, were not tied to one fixed deformation pattern. Instead, they were preserved within a small group of nearby patterns, even when the individual labels inside that group change order between subjects. This matters because it changes how mechanically variable anatomies should be compared: some functions can be studied mode-by-mode, whereas others are described more faithfully at the level of a small subspace. The same logic may help identify robust mechanical function in other biologically variable structures.

1 Introduction

Some biological structures must maintain identical functions across individuals which differ in size, shape, and tissue composition. In load-bearing anatomy, that problem is especially clear as the same biological tasks must be met despite substantial variation in geometry and material distribution. The human pelvis exemplifies this challenge. It must transfer habitual loads between spine and lower limbs while also accommodating variation in inlet and outlet dimensions that is central to locomotor and reproductive biology [1–4]. A central computational question is therefore how pelvic mechanical function remains reliable across anatomically variable individuals.

Robustness in biology is defined by maintenance of function under perturbation rather than rigid preservation of every internal component [5–7]. Degeneracy extends this idea by allowing different internal pathways to support overlapping functions in some contexts while differing in others [8, 9]. For mechanics, this means that transmitting load, opening a cavity, or resisting shear need not rely on exactly the same internal deformation pathway in every individual. Different pelvises may realise a shared function through slightly different low-cost deformation patterns, much as a common motor task can be achieved through a limited but non-unique set of muscle-activation patterns [10].

Population-scale computational biomechanics has difficulty resolving this issue because it often treats ranked deformation pathways as if their indices were biologically stable across subjects. That assumption conflates three different objects: anatomical coordinates that describe shape, mechanical deformation pathways that describe how a structure can move, and spectral order that ranks those pathways by stiffness. These layers need not vary together. In the generalised eigenproblem of linearised elasticity, rank order is meaningful only while neighbouring eigenvalues remain sufficiently separated. When neighbouring eigenvalues approach one another, small perturbations can rotate eigenvectors, exchange labels, and generate veering-like behaviour even when the mechanically relevant span remains intact [11, 12]. In that setting, the stable object is often the *invariant subspace* rather than any individual eigenvector [13]. Comparable spectral behaviour is well known in engineered systems [14, 15], but its biological meaning at population scale remains unclear.

Translated back to biology, the key question is not whether every subject can be forced into one universal rank catalogue. It is whether a function remains attached to an individually stable pathway or is preserved only within a small recurrent family of nearby pathways. Not every invariant subspace is biologically informative: the full modal space is trivially invariant, while arbitrarily large blocks dilute functional specificity. The relevant unit is a compact recurrent subspace, namely a small externally separated block that preserves a targeted function across subjects. At that scale, robustness and degeneracy can be studied without losing anatomical interpretability.

The pelvis provides a biologically meaningful testbed for this question rather than a merely convenient finite-element system. Its gross deformation programs are anatomically interpretable because joint geometry, cartilage, and ligamentous restraints strongly constrain how the ring can move [16–18]. Previous work has used modal finite-element analysis in pelvic and lower-trunk systems [19, 20], and population pelvic anatomy has been summarised through statistical shape models and related representations [4]. What remains missing is a population-scale method that links anatomical variability to mechanical function while keeping distinct the three layers of shape, pathway identity, and spectral rank.

Therefore, the following objective has been studied: under subject-specific bony shape and bone elasticity, which pelvic mechanical functions remain carried by individually stable pathways, and which are preserved only within small recurrent clusters of interchangeable pathways? To answer this, we analyse the linearised stiffness spectrum of 278 subject-specific pelvic models on a shared reference domain. The linear spectrum is used here as a description of the mechanically available deformation repertoire before a particular task is imposed, not as a simulation of contact-rich childbirth. The parturition-motivated load cases are comparative probes of routing [21, 22], and the analysis remains explicitly linear. Across the cohort, we find a gap-protected low-rank backbone that dominates standing loads, recurrent higher-rank clustered regimes in which function can shift between labels without being lost, and clear evidence that the biologically interpretable unit changes from individual mode to compact subspace when spectral gaps collapse.

2 Results

2.1 Interpretable deformation pathways preview two distinct organisational scales

Before turning to cohort statistics, it was investigated whether the low-order deformation repertoire already reveals where mechanical meaning is stably localised and where it is likely to be preserved only at the level of a small clustered family. Figure 1 shows the median deformation patterns for the first fifteen modes in anterior–posterior view. The visualisation immediately suggests a contrast between a compact low-rank program and a richer, more fragile higher-rank repertoire.

Modes 1–3 form the clearest low-rank program. They correspond to the three dominant rotational compliance pathways of the ring: torsion about the craniocaudal axis, sagittal nutation/counternutation about the mediolateral axis, and lateral obliquity about the anteroposterior axis. These are not rigid-body modes in the strict mathematical sense, but they behave as an interpretable low-rank backbone that, as shown below, routes standing-type loads.

Higher modes raise a different biological question: whether a mechanically interpretable function remains tied to one label or only to a compact family. The most important case is the 9–10 pair. Mode 9 typically exhibits an inlet-opening deformation in which the pubic symphysis and the sacral promontory move apart, widening the pelvic ring. Mode 10 is a near-degenerate partner with an asymmetric field concentrated around the pubic region. Their close spacing already suggests that inlet-related behaviour may not be consistently attached to one fixed rank label across subjects, even if it remains preserved within the two-dimensional span they share. For this region of the spectrum, the pair is therefore a more reliable biological interpretive unit than either label alone.

2.2 Population variability separates an order-locked backbone from recurrent clustered regimes

The cohort of $N=278$ subjects resolves the spectrum into two regimes: an order-locked low-rank backbone and recurrent higher-rank clustered blocks (Fig. 2, Table 1). Modes 1–3 are rarely reordered across the cohort (swap rate $< 0.4\%$), whereas higher ranks contain small consecutive gaps and frequent label exchange. Using a data-driven cluster threshold $\varepsilon_{\text{in}} \approx 0.131$ (25th percentile of pooled relative consecutive gaps; see Methods), we detect nine near-degenerate rank clusters above a 3% prevalence floor. The most prevalent cluster is rank modes 9–10 (54.7% of subjects), followed by rank modes 12–15 (41.7%) and rank modes 5–6 (33.8%). No exact degeneracy is observed, as expected for a structure without continuous rotational symmetry.

Anatomical variability does not destabilise the spectrum uniformly. Label fragility is concentrated in a small number of compact families, while low-rank order is preserved. For the dominant 9–10 and 12–15 clusters, median relative internal gaps are small enough that modest perturbations can reverse local ordering. The clusters nevertheless remain externally separated from neighbouring modes, so they are best interpreted as compact clustered subspaces rather than as a global loss of spectral organisation. This pattern is consistent with a pelvic system that preserves stable baseline load-transfer pathways while allowing selected higher-rank functions to vary within a limited local repertoire.

Permutation statistics reinforce this interpretation. Modes 12 and 13 exhibit the highest swap rates, and mode 15 also shows a sizeable unpaired fraction under strict cross-subject pairing, whereas the backbone is essentially fixed in its ordering. For population computational biology the consequence is concrete: ordinary across-subject comparison is safe for the low-rank regime, but higher-rank comparison must accommodate mode mixing and label exchange.

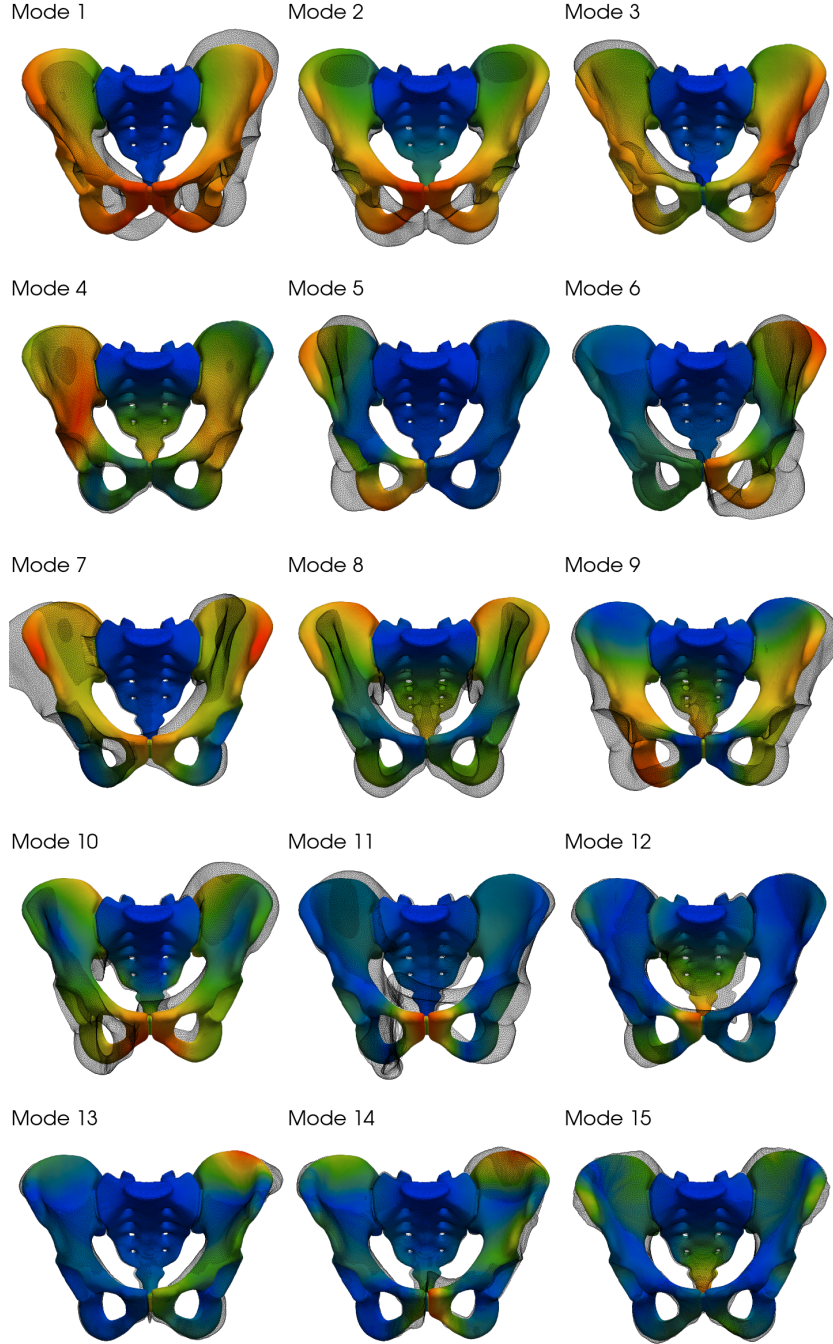


Figure 1: **Interpretable deformation pathways preview the spectrum’s two organisational scales.** Median displacement fields across the cohort ($N=278$) are shown in anterior–posterior view with superimposed deformed wireframes. Colour scale indicates displacement magnitude from blue (minimum/fixed S1 facet) to red (maximum nodal displacement). Modes 1–3 define an order-locked low-rank rotational backbone (craniocaudal torsion, mediolateral nutation, anteroposterior obliquity). Modes 9–10 preview a compact clustered case where inlet opening is preserved at the subspace level rather than by a fixed rank label.

A variability decomposition into shape-only and material-only datasets (Fig. 2) shows that shape variability dominates the coefficient of variation for modes 1–8, whereas bone-density variability becomes more comparable in modes 9–15. The combined variability exceeds either source alone, indicating that shape and density do not simply add independent noise. Shape changes alter *how* the same spatial density heterogeneity enters the stiffness operator. This operator-level interaction between anatomy and material is exactly the kind of reweighting that can preserve a low-rank backbone while pushing selected higher-rank directions toward clustered behaviour.

These prevalence estimates are not driven by a few outliers or by the order in which subjects are processed. Bootstrap resampling of random 80% subcohorts yields narrow 95% subsampling stability intervals for the dominant clusters (9–10: 51.8–57.7%; 12–15: 38.7–45.0%), and full subject bootstrap with replacement confirms consistent prevalence estimates. Random permutation of subject order leaves cluster prevalence and swap statistics unchanged to machine precision. The central spectral hierarchy is therefore a cohort-level feature rather than a post-processing artefact. Additional cluster-mapping, swap, and robustness diagnostics are provided in Appendix A and Appendix B.

Table 1: **Recurrent clustered rank blocks detected above the 3% prevalence floor.**

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	0.4331	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	0.1729	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	0.1588	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

2.3 Clustered subspaces preserve distinct mechanical programs under variability

Once individual labels become unstable, the next question is whether the corresponding small recurrent subspaces remain mechanically distinct. Figure 3 shows that several do, when interpreted using invariant-subspace descriptors rather than individual eigenvector labels. Two basis-invariant outputs were evaluated. First, Grassmann distance from a reference subject quantifies how much a subspace rotates across the cohort. Second, canonical coupling to anatomical diameter functionals (AP and ML inlet; biischadic and bituberous outlet) provides functionally interpretable coordinates inside each clustered block (Methods).

The rank 9–10 subspace shows the strongest inlet-related coupling among the prevalent clusters. Its median AP-inlet coupling is 0.52 mm of inlet diameter change per 1 mm of maximum nodal modal amplitude, making 9–10 the clearest higher-rank inlet-related program in the cohort. In contrast, the rank 5–6 cluster shows the strongest outlet-related coupling, with median biischadic widening 1.41 mm/mm and a smaller bituberous response (0.24 mm/mm). Different clustered subspaces therefore carry different mechanical programs: one is functionally aligned with the inlet, another with the outlet.

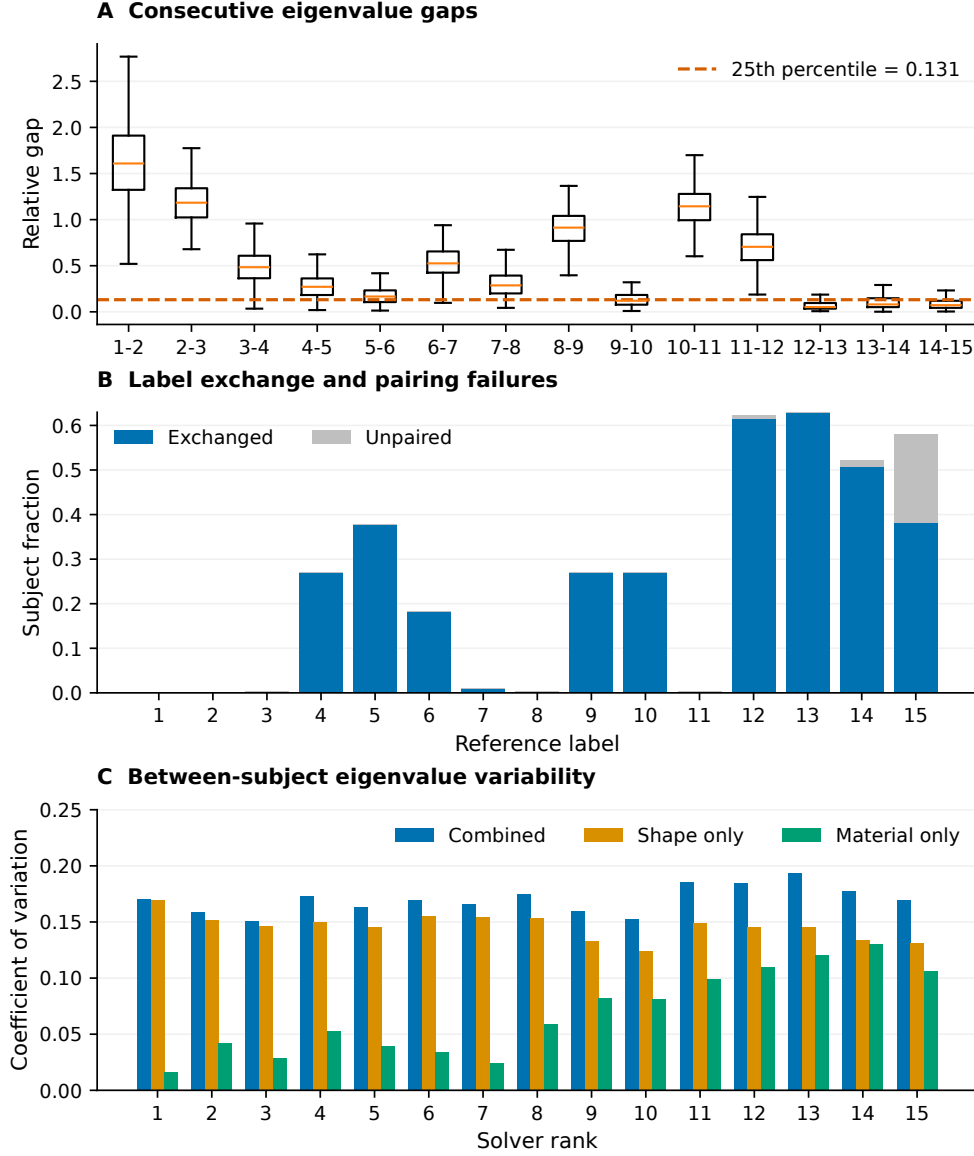


Figure 2: **Population variability partitions the stiffness spectrum into an order-locked backbone and clustered higher ranks.** (A) Consecutive relative spectral gaps $\text{gap}_{\text{in}}(i) = (\lambda_{i+1} - \lambda_i)/\lambda_i$ across the cohort ($N=278$); the dashed line marks the data-driven cluster threshold $\varepsilon_{\text{in}} \approx 0.131$ (25th percentile). (B) Subject fraction exhibiting label exchange (blue) and unpaired pairing failures (grey; $\text{MAC}_{M_0} < 0.05$) per reference label. (C) Eigenvalue variability (coefficient of variation, CoV) across the combined cohort (blue), shape-only cohort (orange), and material-only cohort (green).

The four-dimensional 12–15 block illustrates a third regime. It has high prevalence, small internal gaps, and a richer pattern of within-block mixing than a simple two-mode exchange. Its canonical coupling structure is more diffuse than in 9–10 or 5–6, consistent with it representing a broader, higher-order reserve rather than a single sharply interpretable program. Not every recurrent cluster corresponds to one clean anatomical function. This heterogeneity is itself biologically informative: the framework separates compact, functionally specific blocks from mixed higher-rank

families, and the former are the clearest targets for single-function biological interpretation.

Subspace rotation also varies systematically with sex. The largest such effect is seen for the 9–10 block, suggesting that subjects can reach the same inlet-related subspace through different basis orientations. This qualifies how clustered behaviour is distributed across the population without altering the central point: recurrent clustered subspaces can retain distinct mechanical meanings even when their internal basis orientation varies.

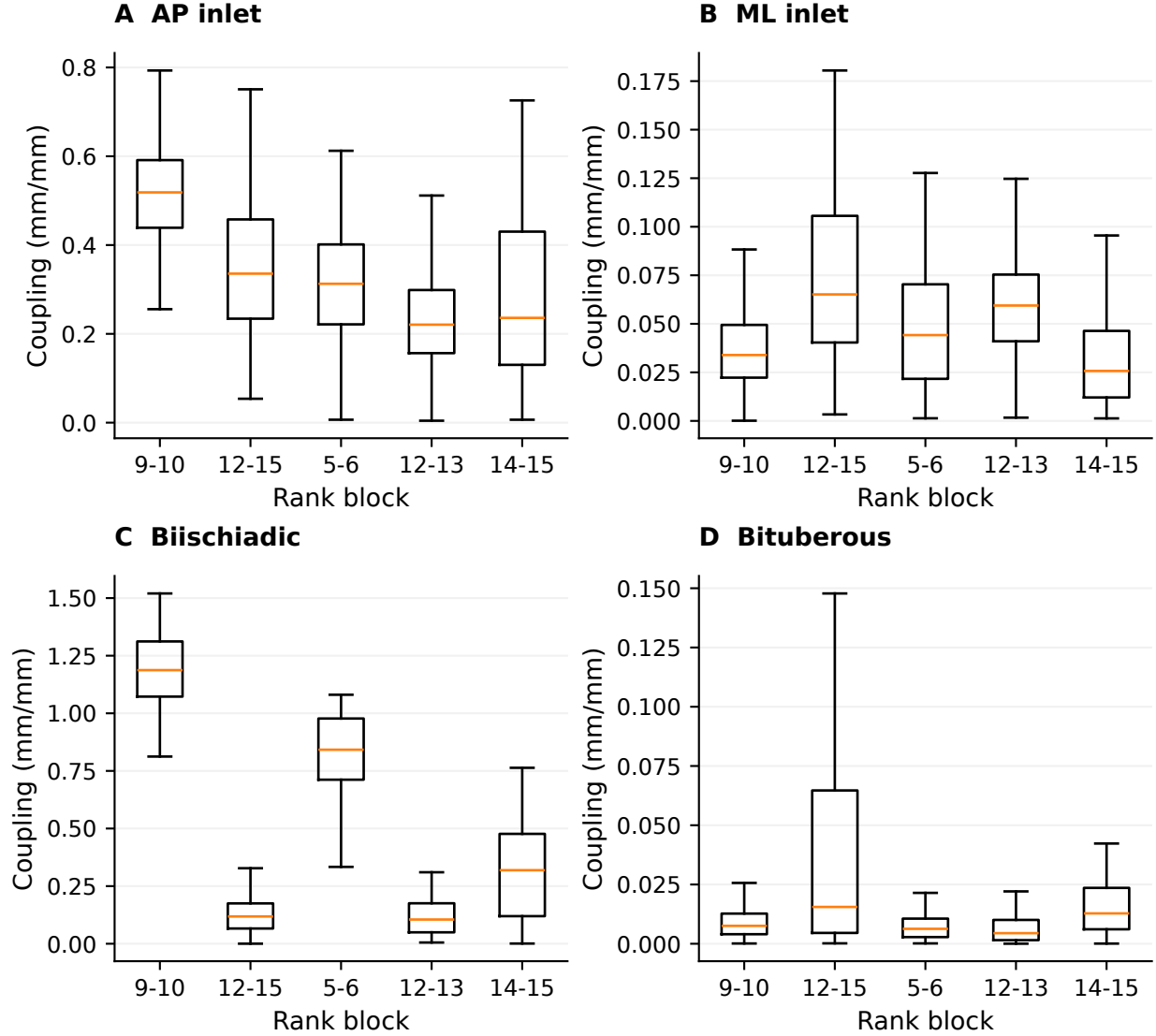


Figure 3: **Anatomical diameter coupling across recurrent clustered modal subspaces.** Canonical coupling scores (mm diameter change per mm maximum nodal displacement) for the five most prevalent rank blocks across the cohort ($N=278$): (A) anteroposterior (AP) inlet, (B) mediolateral (ML) inlet, (C) biischiadic outlet, and (D) bituberous outlet. Boxes indicate median and interquartile range (IQR); whiskers extend to $1.5 \times \text{IQR}$.

2.4 Subspace-level preservation is more stable than fixed labels in clustered regimes

A direct consequence of this framing follows: in a clustered regime, function should be preserved more stably at the scale of the subspace than at the scale of its individual labels. This hypothesis was tested for the most prevalent cluster, rank modes 9–10 (Fig. 4).

Among subjects within the 9–10 cluster, AP-inlet coupling attached to *rank mode 9* is high when labels do not swap and low when they do (median 0.454 vs. 0.139 mm/mm; difference -0.315 mm/mm, 95% CI $[-0.360, -0.262]$, $p = 2.7 \times 10^{-23}$, rank-biserial $r_{rb} = -0.97$). The complementary pattern appears for *rank mode 10* (median 0.163 vs. 0.436 mm/mm; difference $+0.273$ mm/mm, 95% CI $[0.225, 0.335]$, $p = 1.7 \times 10^{-22}$, $r_{rb} = 0.95$). Label-level coupling therefore redistributes between the two rank indices, not because the inlet-related signal disappears, but because it shifts inside the same two-dimensional block while the solver labels exchange.

By contrast, the *subspace-level* AP coupling of the 9–10 block remains substantial in both groups and shifts only modestly (median 0.523 vs. 0.509 mm/mm; difference -0.014 mm/mm, 95% CI $[-0.074, 0.012]$, ratio 0.97, $p = 0.083$, $r_{rb} = -0.17$). What becomes unstable is the attachment of the function to one rank index rather than another, not the function itself. Across-subject comparisons of higher-order clustered modes phrased as though “mode 9” or “mode 10” were fixed entities across the cohort can therefore be misleading. For this regime, the biologically meaningful unit is a compact recurrent subspace—here the shared 9–10 block.

Load routing reinforces the same conclusion (Fig. 5). The 9–10 exchange is primarily a labelling instability with quantitative tuning, not a loss of the inlet-related program. Subspace rotation and the internal 9–10 gap vary across subjects, AP coupling shifts modestly with swap status, and the throughput of the 9–10 block under LAB₁–LAB₃ depends on morphology in a phase-specific way. For LAB₁, block throughput decreases with larger AP inlet diameter ($\rho_s = -0.48$, $p = 4.3 \times 10^{-17}$), whereas LAB₂ and LAB₃ show weaker and differently signed trends. These patterns support the interpretation that inlet-related mechanics remains associated with the 9–10 subspace even as anatomical variability redistributes its basis representation and modulates its phase-specific amplitude.

2.5 Standing and proxy loads distinguish backbone-dominated routing from higher-rank recruitment

Load routing reflects the same separation between conserved baseline support and selective higher-rank recruitment (Fig. 6, Table 2). Standing is dominated by the gap-protected low-rank backbone. Symmetric two-leg standing (SP2leg) is concentrated in mode 2 alone, which carries about 84.5% of the retained modal strain energy; single-leg standing (SP1leg) distributes energy across the full rotational triad (modes 1–3), reaching a top-3 cumulative share of about 92.1%. Within this linear setting, habitual load transfer therefore operates through a small, stable, low-dimensional program.

The proxy load family recruits higher-rank modes. LAB₁, bilateral mediolateral ring compression, preferentially engages the rank 9–10 inlet-related subspace, which carries a mean energy share of about 0.51 and is led by the inlet-opening mode. LAB₂, bilateral ischial-tuberosity distraction, reroutes energy into the mid-rank block at modes 4–8, where the 5–6 outlet-related pair becomes especially important. LAB₃, anteroposterior outlet distraction, is hybrid, retaining sizeable low-rank contributions while also recruiting mid-rank and 9–10 pathways. Different loading geometries recruit different parts of the higher-rank spectrum, while continuing to use the backbone as common support.

Among the recurrent clustered regimes, 9–10 remains the clearest case because it combines high

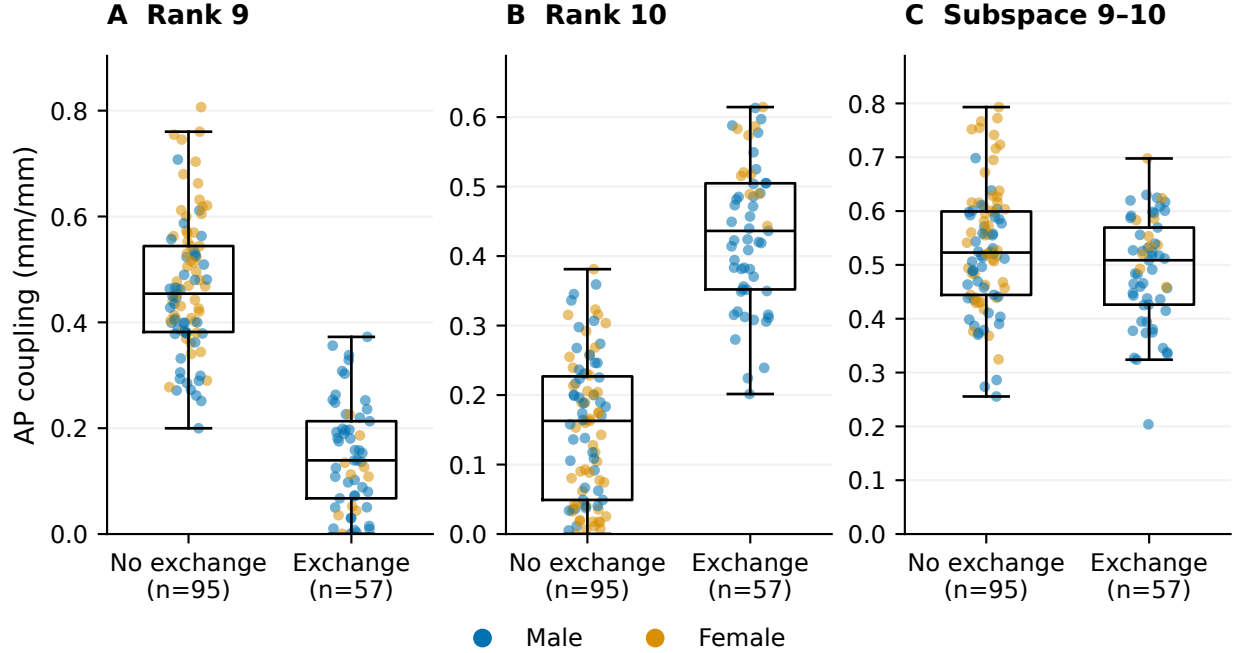


Figure 4: **Inlet coupling redistributes between rank labels within the stable 9–10 subspace.** Anteroposterior (AP) inlet coupling for rank mode 9 (A), rank mode 10 (B), and the canonical two-dimensional 9–10 subspace (C), stratified by $9 \rightleftharpoons 10$ label-exchange status. Points represent individual subjects (blue: male, $n=95$; yellow: female, $n=57$ across cluster members). Horizontal lines indicate medians; boxes indicate interquartile range; whiskers indicate $1.5 \times \text{IQR}$. Coupling is expressed as diameter change per millimetre of maximum nodal displacement.

cohort prevalence, clear inlet-related interpretation, direct label-versus-subspace evidence (Fig. 4), and substantial recruitment under LAB_1 . The 5–6 pair remains an outlet-related secondary contrast under LAB_2 , whereas 12–15 is common but behaves, under the present load family, as a mixed higher-rank reserve rather than a single dominant pathway. Additional load-specificity and sex-stratified summaries are reported in Appendix B and Appendix C.

These are linear comparative proxy loads, not simulations of labour. They do not model fetal contact, tissue remodelling, hormone-dependent softening, or nonlinear late-stage birth mechanics. Their value is comparative: they reveal which deformation programs are preferentially recruited under different loading geometries, not whether the model can predict clinical childbirth outcomes.

Table 2: **Dominant mode sets for the standing and proxy loading cases.** N_{80} denotes the smallest set of modes required to reach 80% cumulative mean energy fraction.

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB_1	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB_2	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB_3	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

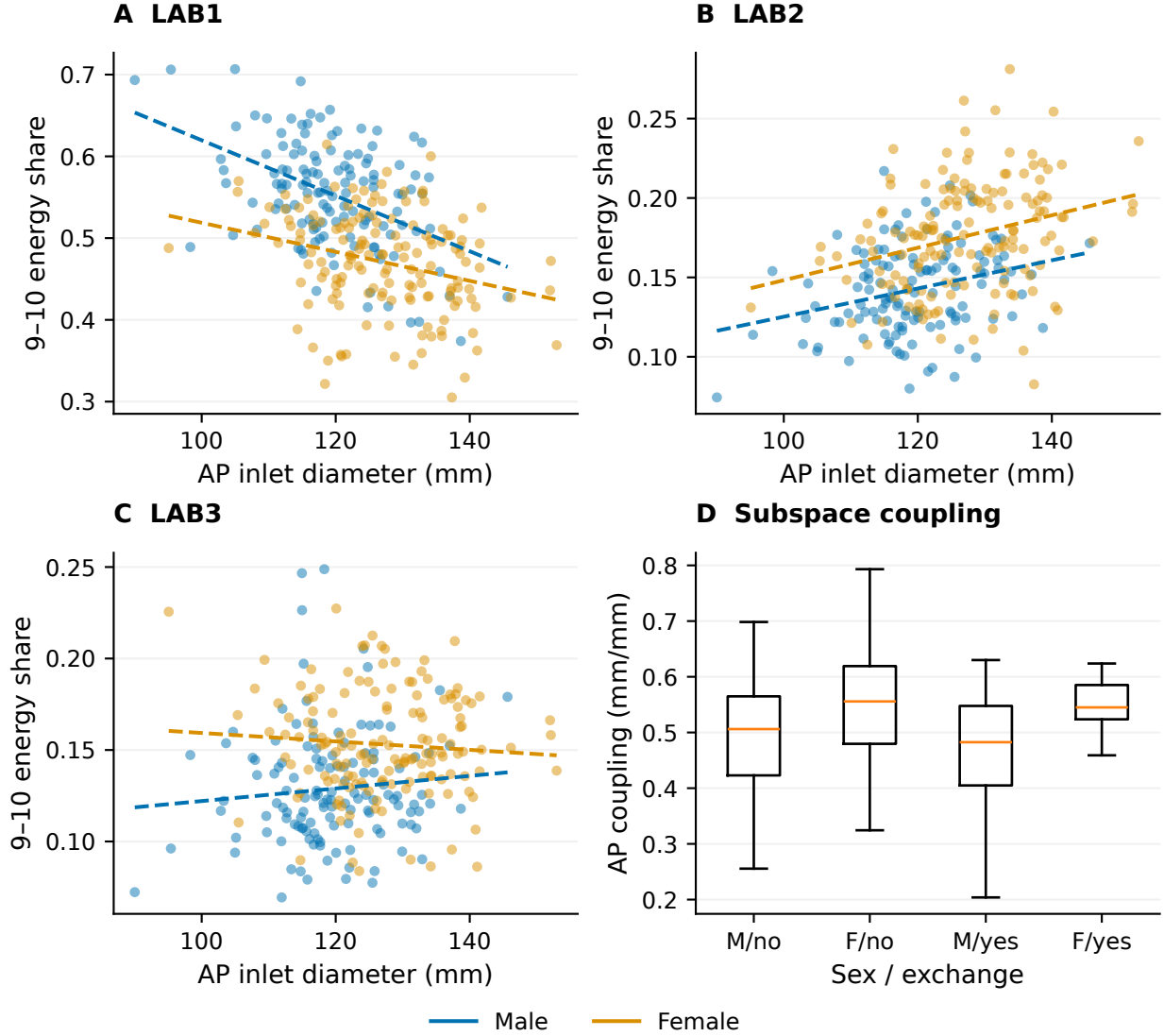


Figure 5: **Morphology and loading modulate recruitment of the 9–10 modal block.** (A–C) Modal strain-energy fraction carried by the 9–10 block as a function of AP inlet diameter for LAB₁ (bilateral ring compression), LAB₂ (ischial distraction), and LAB₃ (outlet distraction). Blue dots and dashed lines indicate males; yellow dots and dashed lines indicate females. (D) Canonical AP-inlet coupling of the 9–10 subspace stratified jointly by sex and label-exchange status (M/no, F/no, M/yes, F/yes). Boxes show median and IQR; whiskers extend to $1.5 \times \text{IQR}$.

2.6 Load-routing overlap distinguishes shared backbone support from selective clustered recruitment

Overlap analysis confirms that shared low-rank support and task-specific higher-rank routing are not the same thing. Both standing cases remain backbone-dominated, yet they are not identical: SP2leg reaches 80% cumulative throughput with mode 2 alone, whereas SP1leg requires the full triad modes 1–3. Among the proxy loads, LAB₁ is distinguished by its emphasis on the 9–10 inlet-related subspace, whereas LAB₂ and LAB₃ share greater mid-rank content, which clarifies why

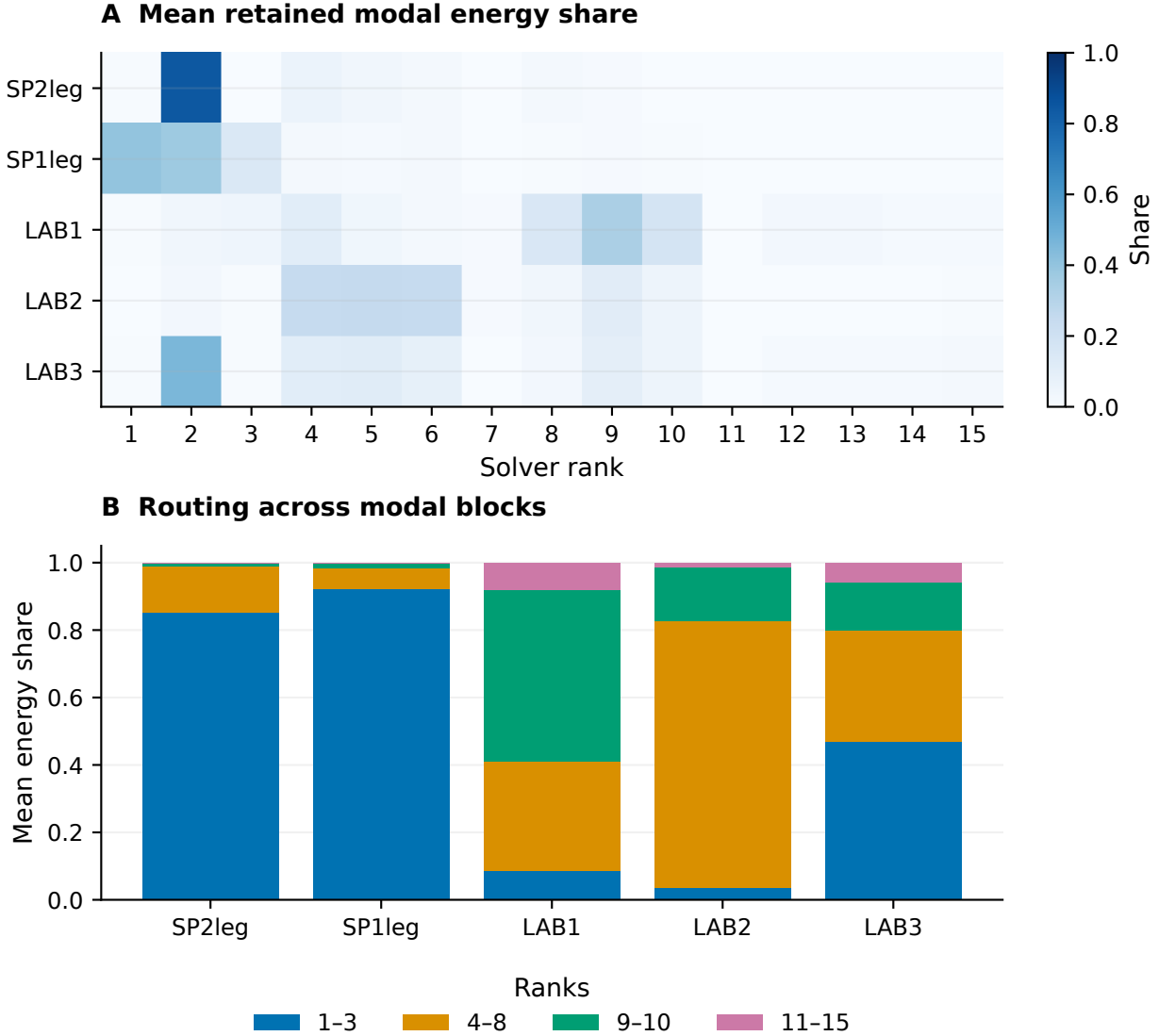


Figure 6: **Standing and proxy childbirth loads recruit distinct modal regimes.** (A) Heatmap of mean retained modal strain-energy share across the 15 lowest solver ranks for standing (SP2leg, SP1leg) and proxy childbirth loads (LAB₁–LAB₃). (B) Partition of retained modal strain energy across functional blocks: low-rank backbone (modes 1–3; blue), mid-rank regime (modes 4–8; orange), inlet cluster (modes 9–10; green), and higher-rank reserve (modes 11–15; purple). Standing concentrates in the backbone, whereas proxy loads recruit higher ranks in a task-specific pattern.

the 5–6 pair is an outlet-related secondary result rather than a side observation. Shared dominant modes and shared full routing profiles are therefore complementary but inequivalent descriptors. Population-scale spectral structure thus separates a conserved low-rank backbone from a selectively recruited higher-rank repertoire whose function is typically more stable at the subspace level than at the level of individual labels. Detailed pairwise overlap values are provided in Appendix C.

2.7 A minimal two-mode model reproduces the observed label–subspace dissociation

The organisation observed in the pelvic cohort admits a compact analytical interpretation in terms of the two- and three-degree-of-freedom models of Sec. 4.12. Fig. 7 shows a stochastic cohort in which a morphology coordinate μ modulates the detuning of a clustered pair on top of a gap-protected backbone, with additional cohort-level nuisance variability introduced through a compliant hidden pathway of scale η . The three model behaviours mirror the three empirical findings of the preceding subsections. First, the backbone mode keeps a stable rank across all subjects and all values of μ (Fig. 7A), matching the order-locked behaviour of modes 1–3 in the FE cohort. Second, as μ crosses the near-degenerate zone, the two cluster labels exchange their coupling to an inlet-like functional while the two-dimensional cluster subspace continues to carry the function (Fig. 7B), reproducing the label-versus-subspace split documented for the 9–10 pair (Fig. 4). Third, a standing-like load routes almost entirely through the backbone, while a task-like load recruits the clustered block, with the label-level split between its two modes moving between subjects but the block-level recruitment staying high (Fig. 7C), mirroring the backbone-versus-higher-rank contrast in the SP2leg/SP1leg and LAB₁–LAB₃ load cases. Increasing the hidden-pathway variability scale η at fixed morphology broadens the label-swap rate while leaving subspace coupling and task routing essentially unchanged (Fig. 7D), illustrating how cohort-level nuisance variability expected from unmodelled tissue heterogeneity can change which mode carries the function without changing whether the function is carried. The two-dimensional cluster in the toy model plays the same role as $\text{span}(Q_s)$ in the FE framework, and the morphology coordinate μ plays the role of the empirical AP-inlet diameter: both are gentle anatomical parameters that can drive a well-defined functional pair across the near-degenerate zone without abolishing its biological content. We note that a permutation null test across the empirical cohort did not show statistically significant swap localization to AP diameter alone (Table A6), indicating that the toy model serves as a mechanistic illustration of how parameter-driven detuning across a near-degenerate zone generates veering, rather than as a proof of one-dimensional anatomical causality. This minimal model therefore formalises why the interpretive unit must shift from *mode label* to *cluster subspace* precisely in the regime in which gaps collapse, and why this shift does not imply loss of mechanical specificity.

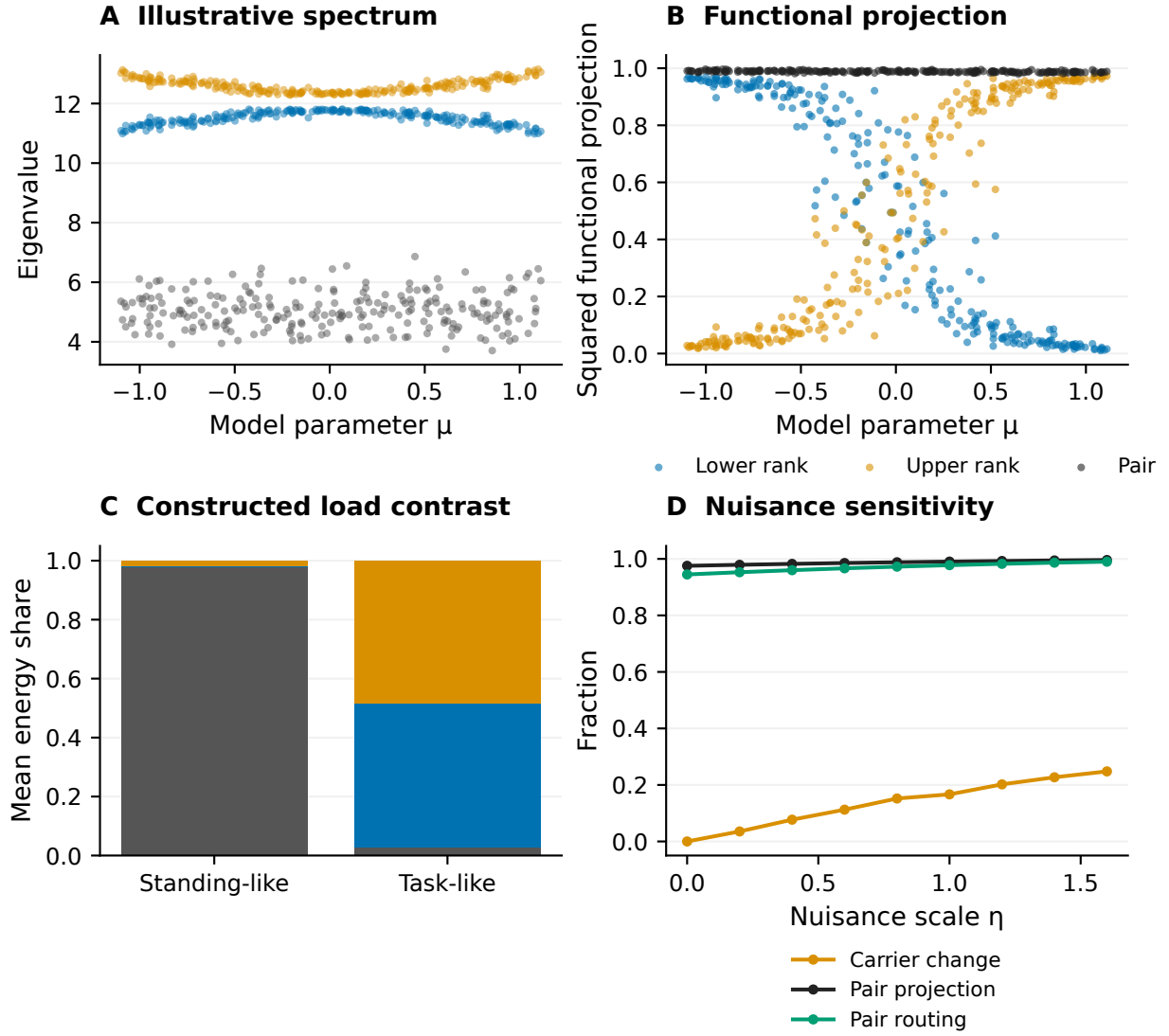


Figure 7: Minimal analytical model illustrates label exchange within a stable subspace. A stochastic cohort of the analytical model with morphology parameter μ and hidden-pathway scale η . **(A)** Spectrum: a separated backbone mode and a nearby eigenvalue pair undergoing near-degeneracy around $\mu \approx 0$. **(B)** Functional projection: coupling shifts between individual rank labels across the transition zone while total coupling within the pair remains high. **(C)** Load routing: standing-like load concentrates in the backbone, whereas task-like load recruits the clustered block with a variable between-label split. **(D)** Effect of hidden-pathway variability scale η : label-exchange rate increases with η , whereas subspace coupling and task routing remain stable. Model derivations and simulation parameters are detailed in Methods.

3 Discussion

Subspace preservation resolves the tension between anatomical variability and functional stability

This given pelvic cohort shows that mechanical function can remain robust across substantial anatomical variability without preserving a one-to-one mapping between function and spectral rank. In this dataset, habitual load transfer is concentrated in a low-rank backbone, whereas selected higher-rank functions are preserved only at the level of compact clustered subspaces. The biological point is not simply that mode labels can become unstable. It is that anatomical variation can redistribute function within a small local repertoire while leaving the function itself available.

This distinction separates three objects which are often treated as if they were interchangeable: anatomical shape, deformation pathway, and spectral order. Shape can vary without preserving a label, and labels can exchange without erasing a function. In musculoskeletal biomechanics, a classic challenge is explaining how structures with pronounced morphological divergence can perform identical kinematic and load-bearing tasks across a population. When analysis is restricted to fixed eigenvector indices, subject-to-subject variability can easily be mistaken for functional degradation or chaotic loss of organisation. By viewing the system through the lens of invariant modal subspaces, we demonstrate that mechanical meaning is not lost; rather, the appropriate unit of biological organisation shifts from the isolated eigenmode to a compact, recurrent subspace. For population studies, the first question is therefore whether a function is order-locked or cluster-preserved. Only after that distinction is made is it sensible to interpret a given spectral label biologically.

Functional organization across spectral scales: Conserved backbone versus flexible higher-rank reserves

The low-rank backbone represents the order-locked regime. Modes 1–3 remain spectrally separated, rarely reorder across the cohort ($< 0.4\%$ swap rate), and dominate habitual standing loads (carrying, for instance, 84.5% of retained modal strain energy in symmetric two-leg stance). Within the present linear framework, this supports the view that the primary kinematic pathways of the pelvic ring—torsion about the craniocaudal axis, sagittal nutation/counternutation, and lateral obliquity—are governed by a conserved, low-cost baseline program. We do not infer an evolutionary or optimisation principle from these data alone, but the result is consistent with biological robustness being concentrated in a compact, gap-protected foundation [6, 8].

The higher ranks reveal an entirely different organisation characterised by recurrent clustered blocks. The rank 9–10 pair exemplifies this regime: inlet-related dilation is not reliably attached to a single rank label across the cohort, but the shared two-dimensional subspace retains substantial AP-inlet coupling (0.523 vs. 0.509 mm/mm between non-swapped and swapped groups) and is strongly recruited under parturition-aligned proxy loads (LAB_1). This constitutes direct physical evidence for biological degeneracy in structural mechanics: structurally distinct internal pathways can provide interchangeable functional outputs without requiring exact mathematical multiplicity [8, 9, 23]. In parallel, the 5–6 block preserves a specific outlet-widening program (dominated by biischadic diameter change), whereas the broader 12–15 family behaves as a higher-order reserve with diffuse canonical coupling. Higher-rank pelvic mechanics is therefore not a featureless noise floor; it contains both functionally focused and diffuse subspaces.

Implications for the obstetrical dilemma and population biomechanics

The human pelvic ring has long been studied through the framework of the “obstetrical dilemma” [1, 2, 4], which posits an evolutionary trade-off between the structural rigidity necessary for efficient bipedal locomotion and the internal dimensions required for the birth of large-brained neonates. Our modal analysis provides a mechanistic perspective on how this trade-off is organized dynamically. Locomotor loads (such as standing and single-leg support) are routed almost exclusively through the stiff, order-locked backbone (modes 1–3), shielding the ring from excessive distortion during everyday weight-bearing. Conversely, parturition-motivated loading conditions bypass the backbone and selectively recruit higher-rank clustered modes, specifically the inlet-dilating 9–10 subspace.

Furthermore, our minimal analytical model (Sec. 2.7) demonstrates that cohort-level nuisance variability—such as that arising from unmodelled ligamentous laxity, pelvic floor compliance, or subtle morphological detuning—can easily induce label veering between nearby modes. Crucially, as shown in Fig. 7D, increasing the scale of hidden-pathway variability (η) accelerates the rate of label exchange between individual modes without degrading subspace-level functional coupling or load throughput. This indicates that the pelvic ring can tolerate individual-level variations in soft-tissue stiffness and shape while reliably preserving the capacity for inlet and outlet expansion.

Methodological scope, physiological limitations, and future directions

What generalises from this study is the logic for testing robustness, not the exact pelvic cluster layout. Combining spectral gap structure, subspace comparison via Grassmann distance, anatomically defined functionals, and modal strain-energy routing makes it possible to evaluate whether function is preserved through stable individual pathways or through stable subspaces in any anatomically variable, load-bearing biological system. The specific rank partition observed here should not be transferred uncritically to other skeletal assemblies or altered physiological states, nor should every recurrent cluster be assumed to map cleanly onto a single anatomical degree of freedom.

Several limitations bound the physiological interpretation of these findings. First, the present analysis is linearised, evaluating the initial tangent stiffness operator of the pelvic ring. Child-birth is a complex, contact-driven, large-deformation process accompanied by progressive viscoelastic relaxation and hormone-induced ligament softening (principally via relaxin and progesterone) [17, 21, 22, 24]. The linear modal spectrum characterises the deformation repertoire immediately available before large-scale joint separation occurs; it does not simulate the descent of the fetal head or predict clinical labour outcomes [25]. Second, the cohort was assembled from clinical CT scans with a mean subject age of ≈ 54 years, which may not fully represent the pelvic morphology of primiparous women. Third, while baseline cartilage and ligament stiffnesses were calibrated against literature values, subject-specific ligament properties were not available. Nonetheless, our first-order sensitivity analyses (Appendix D) confirm that realistic material uncertainty raises the overall spectral variability without reorganising the underlying gap architecture or dismantling the clustered regimes. Extending this subspace framework to include geometrically nonlinear deformation and individualised soft-tissue kinematics represents a promising direction for future population biomechanics.

4 Materials and methods

The Methods below state the computational framework in self-contained form. The core objects are (i) a pullback-assembled linear-elasticity problem on a shared reference domain, (ii) a subject-

specific generalised eigenproblem, (iii) a cross-subject pairing on a common mass metric, (iv) a gap-based detector for recurrent clustered regimes, (v) invariant-subspace comparison via principal angles and the Grassmann distance, and (vi) basis-invariant anatomical functionals that supply physically interpretable coordinates inside otherwise ambiguous clustered subspaces. Wherever possible, we use notation that separates three distinct objects: the solver *rank* index i (the i -th smallest eigenvalue for each subject), the paired *reference* label $\pi_s(i)$ (a cross-subject identifier), and the *anatomical* functional ℓ (a linear observable defined on the template mesh).

4.1 Ethics statement

This study analysed a previously published, anonymised imaging-derived dataset and its associated precomputed finite-element repository (BoneDat) [3]. No new human imaging or interventions were performed for this work. According to the BoneDat descriptor, use of the source imaging data was approved by the Ethics Committee of the University Hospital Hradec Kralove (reference 202411IO3P) with waiver of informed consent; we refer readers to the dataset publication for full ethical and governance details.

4.2 Cohort and geometry normalisation

We analysed a precomputed set of $N=278$ CT-derived pelvic finite-element models assembled on a common reference domain ($n_F=150$, $n_M=128$; age 16–91 yr, mean 54 ± 19). No new cohort was recruited. The source repository provides standardised pelvic geometries suitable for in silico analysis [3]. A diffeomorphic registration $\chi_s : \Omega_{\text{ref}} \rightarrow \Omega_s$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and Jacobian $J_s = \det F_s > 0$ maps the template domain Ω_{ref} onto each subject domain Ω_s . The pipeline standardises bulk pelvic size so that the main population analysis isolates shape-driven and material-driven variability rather than absolute scale. Figure 8 summarises the anatomical dimensions, ligament representation, and boundary-condition layout used to define the pelvic models analysed here.

4.3 Pullback-assembled linear elasticity on a common reference domain

Let $\Omega_{\text{ref}} \subset \mathbb{R}^3$ be the template domain and Ω_s the subject domain, related by a fixed diffeomorphic registration $x = \chi_s(X)$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and Jacobian $J_s(X) = \det F_s > 0$. Writing $\chi_s(X) = X + \varphi_s(X)$, so $F_s = I + \nabla_X \varphi_s$, small-strain linear elasticity is solved on Ω_s but integrated on Ω_{ref} through the standard change of variables [25, 26]. For trial and test fields $\hat{\mathbf{u}}, \hat{\mathbf{v}} \in V \subset [H^1(\Omega_{\text{ref}})]^3$, the physical strain tensor at $x = \chi_s(X)$ is

$$\boldsymbol{\varepsilon}_x(\mathbf{u}) = \text{sym}(\nabla_x \mathbf{u}) = \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}), \quad \boldsymbol{\sigma}_x = C_{\text{phys}}(x) : \boldsymbol{\varepsilon}_x(\mathbf{u}), \quad (1)$$

and the pullback-assembled elastic bilinear form reads

$$a(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} \left[C_{\text{phys}}(\chi_s(X)) : \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}) \right] : \text{sym}(\nabla_X \hat{\mathbf{v}} F_s^{-1}) J_s \, dX. \quad (2)$$

Here F_s and J_s arise only from the fixed image registration; no finite-strain mechanics is implied. The mass bilinear form inherits the volume factor,

$$m(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} J_s(X) \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} \, dX, \quad (3)$$

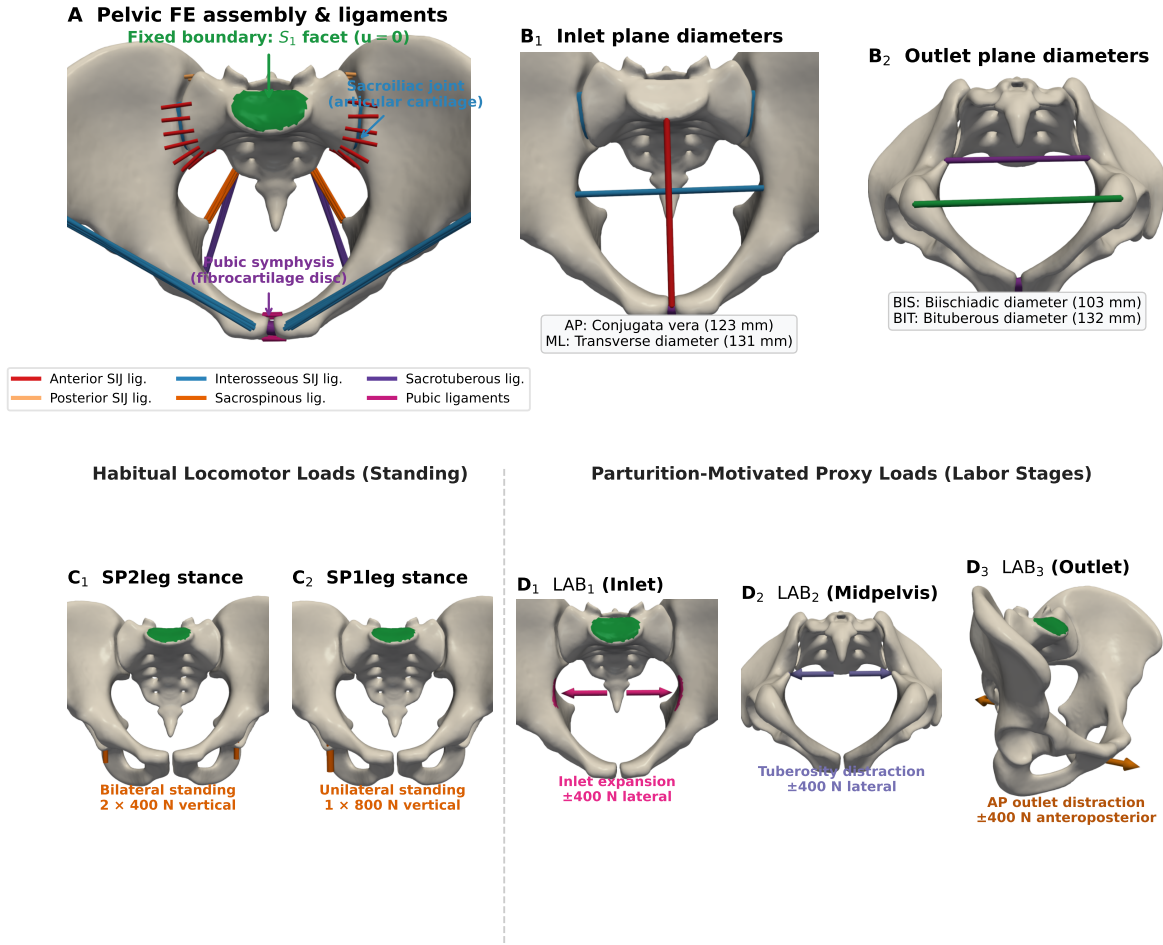


Figure 8: **Pelvic finite-element assembly, anatomical diameters, boundary conditions, and static load configurations.** (A) Complete 3D pelvic finite-element model comprising the sacrum, bilateral ossa coxae, sacroiliac joint (SIJ) articular cartilage (cyan), and pubic symphysis fibrocartilage disc (purple). Eleven discrete anatomical ligament bundles (anterior SIJ, posterior SIJ, interosseous SIJ, sacrospinous, sacrotuberous, and pubic ligaments) are explicitly represented. The superior articular surface of the sacrum (S_1 facet, green) is fixed against rigid-body displacement ($\mathbf{u} = \mathbf{0}$) via surface penalty. (B) True planar normal projections for anatomical pelvimetric diameters: (B₁) pelvic inlet plane showing Conjugata vera (true conjugate, red, 123 mm) and transverse diameter (blue, 131 mm); (B₂) pelvic outlet plane (inferior/perineal projection) showing biischadic (interspinous) diameter (purple, 103 mm) and bituberous diameter (green, 132 mm). (C) Habitual locomotor load cases representing upright stance: (C₁) bilateral standing (SP2leg, 2×400 N vertical joint reaction forces applied at left and right acetabular notches); (C₂) unilateral single-leg stance (SP1leg, 1×800 N vertical joint reaction force applied at the right acetabulum). (D) Parturition-motivated proxy load cases representing stages of fetal transit through the pelvic canal: (D₁) inlet ring expansion during head engagement (LAB₁, ± 400 N lateral expansion forces applied at the iliopectineal inner ring contacts); (D₂) midpelvis transit (LAB₂, ± 400 N lateral distraction forces applied across the ischial tuberosities/spines); (D₃) outlet clearance (LAB₃, ± 400 N anteroposterior distraction consisting of anterior traction on the pubic crest and posterior traction on the sacrococcygeal joint).

whereas the *unmapped* reference mass matrix M_{ref}^0 is obtained from (3) with $J \equiv 1$ and provides a shared inner product across subjects that is independent of any particular subject-specific registration.

Material model. The physical elasticity tensor $C_{\text{phys}}(x)$ is isotropic and heterogeneous. For bone, the CT-derived hydroxyapatite density $\rho(x)$ is converted to ash density $\rho_{\text{ash}} = (\rho + 0.09)/1.14$ and mapped to a Young’s modulus $E(x)$ via the Keller [27] piecewise power law,

$$E(x) = \begin{cases} 2398 \text{ MPa}, & \rho_{\text{ash}}(x) \leq 0.486, \\ \alpha \rho_{\text{ash}}(x)^\beta, & \rho_{\text{ash}}(x) > 0.486, \end{cases} \quad \alpha = 10\,200, \beta = 2.0, \quad (4)$$

with Poisson’s ratio $\nu = 0.3$. This family of density–modulus relations brackets published calibrations spanning the classical Carter–Hayes exponent [28] and the site-dependent regressions of Morgan et al. [29]. Cartilage (sacroiliac joints and pubic symphysis) is homogeneous linear elastic with $E = 1.0 \text{ MPa}$ and $\nu = 0.45$. These choices are consistent with structural adaptation and architecture–stiffness arguments in bone biomechanics [30–33] and with the broader sacroiliac and pubic-symphysis mechanics literature used to motivate linearised pelvic models [16–18].

Ligaments in pullback coordinates. Ligament families are represented as polyline bundles mapped from template nodes X_i to physical points $x_i = \chi_s(X_i)$. For each segment (X_i, X_j) define the physical unit tangent $\mathbf{e} = (x_j - x_i)/\|x_j - x_i\|$ and length $L = \|x_j - x_i\|$. The total bundle stiffness and pretension are distributed across N_{seg} segments, yielding per-segment $k_{\text{seg}} = K_{\text{bundle}}/N_{\text{seg}}$ and $T_{\text{seg}} = T_{\text{bundle}}/N_{\text{seg}}$. Each segment contributes an axial stiffness block

$$W = k_{\text{seg}}(\mathbf{e}\mathbf{e}^\top), \quad K_{ij}^{\text{ax}} = \begin{bmatrix} +W & -W \\ -W & +W \end{bmatrix}, \quad (5)$$

and, in the presence of pretension $T_{\text{seg}} \geq 0$, a geometric (prestress) stiffness

$$K_{ij}^{\text{geo}} = \frac{T_{\text{seg}}}{L}(\mathbf{I} - \mathbf{e}\mathbf{e}^\top) \succeq 0, \quad (6)$$

together with paired nodal forces $\pm T_{\text{seg}}\mathbf{e}$ on the right-hand side. The full assembled stiffness is

$$K_s = K_{\text{elas},s} + K_{\text{lig},s} + K_s^{\text{geo}}(T) + K_{\Gamma,s}, \quad f = f_{\text{ext}} + f_{\text{pret}}(T), \quad (7)$$

where $K_{\Gamma,s}$ implements Dirichlet penalties on Γ_{ref} , $a_\Gamma(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \gamma \int_{\Gamma_{\text{ref}}} \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} J_{\Gamma,s} dS_X$ with $J_{\Gamma,s} = J_s \|F_s^{-T} N\|$ and $\gamma = \alpha_\Gamma \bar{E}/h$, $\alpha_\Gamma \in [10^3, 10^5]$. A decade sweep of γ left the reported cluster classification unchanged.

Force-controlled loading. Neumann terms are pulled back to Γ_{ref} so that physical resultants are invariant under χ_s :

$$\int_{\Gamma_s} \mathbf{t}_{\text{phys}} \cdot \mathbf{v} dS_x = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s}(X) \mathbf{t}_{\text{phys}}(\chi_s(X)) \cdot \hat{\mathbf{v}}(X) dS_X. \quad (8)$$

All load cases in this study are defined by a total force $\mathbf{F}$ on a tagged physical region; the applied traction is $\mathbf{t}_{\text{phys}} = \mathbf{F}/A_{\text{phys}}$ with $A_{\text{phys}} = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s} dS_X$, so that $\int_{\Gamma_s} \mathbf{t}_{\text{phys}} dS_x = \mathbf{F}$ for every subject.

4.4 Standardised load cases

Loads were applied as force-controlled Neumann tractions via (8) so that every subject experienced the same integrated force vector regardless of physical surface area. Two standing cases were considered: symmetric two-leg stance (SP2leg; 400 N per acetabulum, +CC) and one-leg stance (SP1leg;

800 N on the stance acetabulum, +CC). The 400 N per hip in SP2leg is a conservative split of head–arms–trunk mass onto both acetabula based on classical segment-inertia data [34], and the 800 N one-leg value is well below in-vivo hip-implant peaks of $2.5\text{--}3.5 \times \text{BW}$ recorded during single-leg support and gait [35, 36]. Three proxy load cases probed parturition-relevant routing without modelling contact mechanics: LAB₁ applied bilateral mediolateral ring-contact compression (± 400 N), LAB₂ applied bilateral ischial-tuberosity distraction (± 400 N), and LAB₃ applied anteroposterior outlet distraction between the pubis insertion ($-AP$, 400 N) and the sacrococcygeal joint ($+AP$, 400 N). Their order of magnitude is consistent with published summaries of uterine contraction pressures of $\sim 40\text{--}80$ mmHg [37, 38], elevated intra-abdominal pressures during pregnancy [39, 40], and review-level biomechanical estimates of fetal head–canal wall pressures during pushing [24]. These loads are used as comparative probes rather than clinical simulations of labour [21, 22].

4.5 Generalised eigenproblem, rigid-mode filtering, and cross-subject pairing

For each subject s we assembled (K_s, M_s) on Ω_{ref} via (7) and (3) and solved the generalised eigenproblem

$$K_s \phi_k = \lambda_k M_s \phi_k, \quad \phi_i^\top M_s \phi_j = \delta_{ij}, \quad (9)$$

using a Krylov–Schur solver (SLEPc) with shift-invert spectral transform and MUMPS direct factorisation, extracting the $N_{\text{ev}} = 15$ lowest non-rigid eigenpairs. Rigid modes (numerically $\lambda < 10^{-8}$) were filtered. For cross-subject comparison we used the mass-weighted modal assurance criterion on the shared reference metric,

$$\text{MAC}_{M_0}(\mathbf{x}, \mathbf{y}) = \frac{|\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{y}|^2}{(\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{x})(\mathbf{y}^\top M_{\text{ref}}^0 \mathbf{y})} \in [0, 1], \quad (10)$$

and solved the Hungarian optimal-assignment problem on $-\text{MAC}_{M_0}$ over the full $N_{\text{ev}} \times N_{\text{ev}}$ matrix [41], yielding a permutation π_s from reference modes to subject s ’s rank labels. Entries with $\text{MAC}_{M_0} < 0.05$ were flagged as unpaired. Rank index i (solver order) is required for gap statistics and cluster detection; the paired reference label $\pi_s(i_{\text{ref}}) = j_{\text{rank}}$ maps a reference-mode identifier to its rank position in subject s and is used only for anatomical interpretation, preserving the distinction between rank order and pathway identity emphasised in the Introduction.

4.6 Spectral gap structure and clustered-regime detection

For each subject with ordered eigenvalues $\lambda_1 \leq \dots \leq \lambda_{N_{\text{ev}}}$, relative consecutive gaps are

$$\text{gap}_{\text{in}}(i) = \frac{\lambda_{i+1} - \lambda_i}{\lambda_i}, \quad i = 1, \dots, N_{\text{ev}} - 1. \quad (11)$$

A clustered regime $[i..j]$ is defined as the maximal contiguous rank block starting at i with internal gaps below threshold, $\text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ for $k = i, \dots, j - 1$. The threshold is set from data as the 25th percentile of the pooled gap_{in} distribution across the cohort, yielding $\varepsilon_{\text{in}} \approx 0.13$; sensitivity to the percentile choice is reported in Appendix A. External separation is quantified by

$$\text{sep}_{\text{out}}([i..j]) = \min\left(\frac{\lambda_i - \lambda_{i-1}}{\lambda_i}, \frac{\lambda_{j+1} - \lambda_j}{\lambda_j}\right), \quad (12)$$

with a well-isolated cluster satisfying $\text{sep}_{\text{out}} \gg \varepsilon_{\text{in}}$. True degeneracy in the numerical sense ($\text{gap}_{\text{in}} \leq \varepsilon_{\text{deg}} = 10^{-6}$) was not observed in any subject, which is expected for a structure without continuous rotational symmetry; the regimes detected by (11) are genuine near-degenerate clusters, for which small perturbations can nevertheless drive rapid internal basis rotation and label exchange [11, 12].

4.7 Invariant subspaces: principal angles, Grassmann distance, and Procrustes alignment

For each clustered rank block of dimension r , the corresponding subject-specific subspace was represented by an M_{ref}^0 -orthonormal basis

$$Q_s \in \mathbb{R}^{n \times r}, \quad Q_s^\top M_{\text{ref}}^0 Q_s = I_r. \quad (13)$$

With a fixed reference basis Q_{ref} , define the overlap matrix

$$S = Q_{\text{ref}}^\top M_{\text{ref}}^0 Q_s \in \mathbb{R}^{r \times r}, \quad S = U \Sigma V^\top, \quad \cos \theta_k = \sigma_k(S), \quad (14)$$

with singular values σ_k yielding the principal angles $\theta_k \in [0, \pi/2]$ between the two subspaces [42]. The Grassmann distance is

$$d_G(Q_{\text{ref}}, Q_s) = \|\theta\|_2 = \left(\sum_{k=1}^r \theta_k^2 \right)^{1/2}. \quad (15)$$

When a stable basis representation inside a clustered subspace is required, we apply an orthogonal Procrustes rotation [43],

$$\tilde{Q}_s = Q_s (V U^\top), \quad (16)$$

which minimises the Euclidean distance to Q_{ref} while preserving $\text{span}(Q_s)$. All quantitative biological comparisons use basis-invariant scalars derived from (14)–(15) or from the canonical-basis construction below.

Degeneracy vs. near-degenerate mixing. True multiplicity requires $\text{gap}_{\text{in}}(k) \leq \varepsilon_{\text{deg}}$ for all k in the block, with $\varepsilon_{\text{deg}} \ll \varepsilon_{\text{in}}$; individual eigenvectors inside a degenerate subspace are then not unique, and any unitary rotation inside $\text{span}(Q_s)$ is an equally valid basis. When $\varepsilon_{\text{deg}} < \text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ the eigenvalues are close but distinct; the eigenvectors are formally unique, yet small perturbations can rotate the internal basis substantially while $\text{span}(Q_s)$ itself remains nearly preserved. This is the regime in which label exchange occurs without loss of the block-level mechanical content.

4.8 Canonical anatomical functionals inside clustered subspaces

To interpret clustered subspaces biologically we use basis-invariant linear functionals defined by pairs of pelvic landmarks. Let $(X_{\text{ant}}, X_{\text{post}})$ and $(X_{\text{left}}, X_{\text{right}})$ denote the anteroposterior and mediolateral inlet landmarks on the template. Patient-specific unit directions follow the registered positions $X^s = X + \varphi_s(X)$,

$$\mathbf{e}_{\text{AP}}^{(s)} = \frac{X_{\text{ant}}^s - X_{\text{post}}^s}{\|X_{\text{ant}}^s - X_{\text{post}}^s\|}, \quad \mathbf{e}_{\text{ML}}^{(s)} = \frac{X_{\text{right}}^s - X_{\text{left}}^s}{\|X_{\text{right}}^s - X_{\text{left}}^s\|}, \quad (17)$$

and the inlet opening scores for a displacement field $\mathbf{u}$ are the linear functionals

$$s_{\text{AP}}^{(s)}(\mathbf{u}) = \mathbf{e}_{\text{AP}}^{(s)\top} (\mathbf{u}(X_{\text{ant}}) - \mathbf{u}(X_{\text{post}})) = \mathbf{b}_{\text{AP}}^{(s)\top} \mathbf{u}, \quad s_{\text{ML}}^{(s)}(\mathbf{u}) = \mathbf{b}_{\text{ML}}^{(s)\top} \mathbf{u}, \quad (18)$$

with sparse measurement vectors $\mathbf{b}_{\text{AP}}^{(s)}, \mathbf{b}_{\text{ML}}^{(s)} \in \mathbb{R}^n$ supported at the landmark nodes. Outlet scores $s_{\text{BIS}}^{(s)}$ (biischadic) and $s_{\text{BIT}}^{(s)}$ (bituberous) are constructed identically with the corresponding landmark pairs.

Canonical basis inside a clustered subspace. For a pair of measurement vectors ($\mathbf{b}_1^{(s)}, \mathbf{b}_2^{(s)}$) (inlet or outlet) and a clustered subspace $Q_s \in \mathbb{R}^{n \times r}$, project the observables into the block,

$$C_s = [Q_s^\top \mathbf{b}_1^{(s)} \quad Q_s^\top \mathbf{b}_2^{(s)}] \in \mathbb{R}^{r \times 2}, \quad C_s = U_s \Sigma_s V_s^\top, \quad (19)$$

and define the canonical basis

$$\psi_1^{(s)} = Q_s U_s[:, 1], \quad \psi_2^{(s)} = Q_s U_s[:, 2], \quad (20)$$

oriented by the sign convention $\mathbf{b}_1^{(s)\top} \psi_1^{(s)} \geq 0$. The singular values $\sigma_1 \geq \sigma_2$ quantify how strongly the subspace couples to the chosen diameter axes. Because C_s depends on Q_s only through $\text{span}(Q_s)$, the canonical basis is invariant to arbitrary rotations within the subspace, providing a physically stable coordinate system even when individual eigenvector labels are ambiguous. Canonical coupling scores are reported in max-norm-normalised units of diameter change per 1 mm of maximum nodal amplitude,

$$\hat{s}_k^{(s)} = \frac{|\mathbf{b}_k^{(s)\top} \psi_k^{(s)}|}{\|\psi_k^{(s)}\|_\infty}, \quad (21)$$

so that reported couplings are independent of arbitrary eigenvector normalisation. The direct 9–10 label-versus-subspace analysis evaluates AP-inlet coupling for rank mode 9, rank mode 10, and the canonical 9–10 subspace after splitting cluster members by 9 $\rightleftharpoons$ 10 swap status.

4.9 Load routing and modal strain-energy fractions

For each load case we solved the static problem $K_s \mathbf{u}_s = \mathbf{b}$ on Ω_{ref} , projected the result onto the M_s -orthonormal eigenbasis Φ ,

$$q = \Phi^\top M_s \mathbf{u}_s, \quad E_i = \frac{1}{2} \lambda_i q_i^2, \quad f_i = \frac{E_i}{\sum_j E_j}, \quad (22)$$

and summarised each pathway's contribution by its modal strain-energy fraction f_i . Block throughput for a clustered regime C is $f_C = \sum_{i \in C} f_i$, and the dominant mode set N_{80} of a load case is the smallest rank-ordered set whose mean energy fractions sum to 0.8. For cross-load comparison we also report cosine similarity of full 15-mode mean profiles and Jaccard overlap of dominant mode sets.

4.10 First-order eigenvalue sensitivities and analytical uncertainty propagation

To propagate material-parameter uncertainty to the spectrum without repeated finite-element solves, we use classical first-order sensitivities of the generalised eigenproblem (9): for a parameter p_j that affects only the stiffness,

$$\frac{\partial \lambda_m}{\partial p_j} = \frac{\phi_m^\top (\partial K_s / \partial p_j) \phi_m}{\phi_m^\top M_s \phi_m}. \quad (23)$$

For ligament parameters, $\partial K_s / \partial k_{\text{seg}} = \mathbf{e} \mathbf{e}^\top$ and $\partial K_s / \partial T_{\text{seg}} = (I - \mathbf{e} \mathbf{e}^\top) / L$ are aggregated over segments in each bundle using the geometric assembly underlying (6). For cartilage moduli, $\partial K_s / \partial E_{\text{region}}$ is the element-wise elastic stiffness integrated over the corresponding joint domain, scaled by the baseline modulus. For the high-density branch of (4), the chain rule gives $\partial E / \partial \alpha = E / \alpha$ and $\partial E / \partial \beta = E \ln \rho_{\text{ash}}$, integrated over the bone domain. Storing the dimensionless Jacobians

$$J_j^{(m)} = \frac{p_j}{\lambda_m} \frac{\partial \lambda_m}{\partial p_j} \quad (24)$$

once per subject lets us compute log-space variance propagation as $\text{Var}(\log \lambda_m) = \mathbf{J}_m^\top \Sigma \mathbf{J}_m$ for any prior covariance Σ on the material parameters, including hierarchical priors that separate global, per-type, and left/right components. For adjacent eigenvalues λ_i, λ_{i+1} , the variance of the gap $\Delta\lambda = \lambda_{i+1} - \lambda_i$ is computed from the combined sensitivity vector $\mathbf{g}_{i,i+1} = \lambda_{i+1}\mathbf{J}_{i+1} - \lambda_i\mathbf{J}_i$ as

$$\text{Var}(\lambda_{i+1} - \lambda_i) = \mathbf{g}_{i,i+1}^\top \Sigma \mathbf{g}_{i,i+1} = \text{Var}(\lambda_{i+1}) + \text{Var}(\lambda_i) - 2 \text{Cov}(\lambda_i, \lambda_{i+1}), \quad (25)$$

accounting for the substantial positive covariance induced by shared tissue parameters. Fig. 9 summarises the resulting variance decomposition across the 15 lowest eigenvalues and the covariance-aware gap signal-to-noise ratio across solver ranks. The gap-protected low-rank modes and the 9–10 inlet cluster are covered by the same analytical machinery, so the cluster-detection threshold ε_{in} can be compared directly against the propagated spectral uncertainty; cluster boundaries remain well-defined under the posited parameter covariances. Full hierarchical priors, Shapley-style decompositions, and sex-stratified versions are reported in Appendix B and Appendix D.

4.11 Decomposition of variability sources and robustness analyses

To separate shape and material contributions to spectral variability, we compared the full cohort against a shape-only cohort (χ_s , fixed reference bone modulus) and a material-only cohort (identity deformation, subject-specific bone modulus). Robustness was assessed by (i) random permutation of subject order (invariance to machine precision), (ii) subject bootstrap with replacement ($N=278$, 500 replicates) with threshold re-estimation, alongside random 80% subsampling for cohort stability, and (iii) morphology-axis permutation null tests in which a one-dimensional anatomical proxy μ (e.g. AP inlet diameter) was shuffled 1000 times to assess whether swap-rate localisation along μ exceeded chance (Table A6). Post-hoc allometric regressions of $\log \lambda_k$ on isotropic body-size scale s (with instrumental variables from volume, surface, and mass) and Shapley/LMG variance decomposition into scale, sex, shape, and material contributions are reported in Appendix D.

4.12 Minimal analytical model for the near-degenerate regime

The cluster/subspace framework admits a compact analytical interpretation via a two-degree-of-freedom model. Let q_1, q_2 denote two generalised rotational coordinates and define a stiffness matrix

$$K_2 = \begin{bmatrix} k_1 & c \\ c & k_2 \end{bmatrix}, \quad \Delta = k_1 - k_2, \quad k_1, k_2 > 0, \quad k_1 k_2 > c^2, \quad (26)$$

with detuning Δ and coupling c . The eigenvalues are

$$\lambda_{\pm} = \frac{k_1 + k_2}{2} \pm \sqrt{\left(\frac{\Delta}{2}\right)^2 + c^2}, \quad \tan(2\theta) = \frac{2c}{\Delta}, \quad (27)$$

and the minimum separation $\lambda_+ - \lambda_- = 2|c|$ is attained at $\Delta = 0$, where the mixing angle passes through $\pi/4$ and the modes exchange character. The operational criterion

$$|\Delta(\mu)| \leq \tau |c(\mu)| \quad \text{for some morphology parameter } \mu, \quad \tau \in [1.5, 2], \quad (28)$$

separates near-degenerate/mixing from stable-ordering regimes. Adding a third coordinate q_3 with stiffness s and couplings d_1, d_2 and eliminating it by Schur complement yields an effective two-mode system

$$K_{\text{eff}} = \begin{bmatrix} k_1 - d_1^2/s & c - d_1 d_2/s \\ c - d_1 d_2/s & k_2 - d_2^2/s \end{bmatrix}, \quad (29)$$

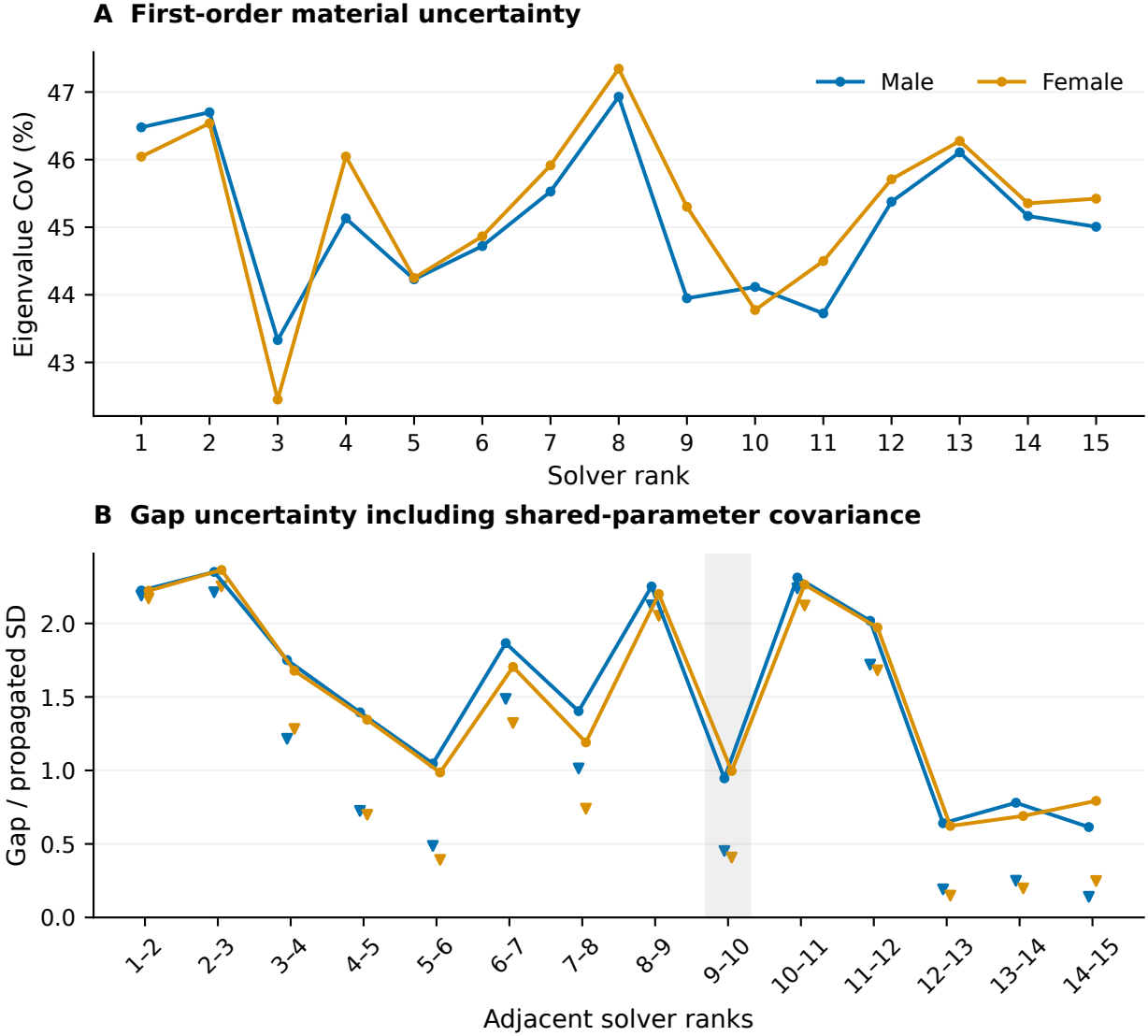


Figure 9: **Analytical propagation of material-parameter uncertainty to eigenvalues and spectral gaps.** (A) Total coefficient of variation (CoV, %) of eigenvalues $\lambda_1, \dots, \lambda_{15}$ under hierarchical material-parameter covariance, stratified by sex (blue: male; yellow: female). (B) Relative gap signal-to-noise ratio (spectral gap divided by propagated standard deviation) between adjacent solver ranks, incorporating shared-parameter covariance $\text{Cov}(\lambda_i, \lambda_{i+1})$. The shaded region highlights the 9–10 rank transition. Triangles mark 10th-percentile values across subjects.

showing how a compliant hidden pathway (small s) can shift effective detuning and coupling and push a population that is stable in the bare two-mode picture toward the near-degenerate regime. In the n -DOF finite-element setting, the detected clusters correspond to $\{q_1, q_2\}$, the canonical basis (ψ_1, ψ_2) to the rotated eigenvectors, and gap_{in} to $(\lambda_+ - \lambda_-)/\lambda_-$. A stochastic population instantiation of this model, shown and discussed in Fig. 7 and Sec. 2.7, reproduces the label–subspace dissociation, the standing-vs-task routing contrast, and the effect of hidden-pathway variability observed in the full pelvic cohort. Explicit derivations, the small- c expansion, and the full stochastic population extension are provided in Appendix D.

4.13 Software and computational implementation

Finite-element assembly, eigensolves, and postprocessing were implemented in Python 3.11 with DOLFINx/FEniCSx (≥ 0.10), PETSc, and SLEPc. All reported figures, tables, and numerical values are produced by scripts included in the study repository. Results are stored as compressed Zarr arrays for reproducible downstream analysis. Further implementation details, dependency versions, and reproduction instructions are provided in the appendices and the code repository.

Data availability

Derived summary data supporting the findings of this study are provided within the manuscript, the appendices, and the study repository. The standardised pelvic geometries originate from the BoneDat research dataset [3]; access to the underlying clinical imaging data remains subject to source-data governance restrictions and requires approval through the procedures described in the dataset publication.

Code availability

Custom code used to assemble the models, run the spectral analyses, and generate the reported figures and tables is available in the study repository. A public archival release of the analysis code will accompany publication.

Author contributions

P.H. and N.H. conceptualised the study. P.H. developed the computational models, performed the simulations, and analysed the data. P.H. and N.H. interpreted the results and wrote the manuscript.

Competing interests

The authors declare no competing interests.

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We thank the BoneDat project for curating the standardised imaging-derived dataset used in this study.

A Spectral cluster architecture

The appendices collect robustness, specificity, and model-boundary analyses that support the main manuscript. They are integrated directly into this manuscript so that the working draft remains self-contained.

This appendix expands the bookkeeping behind the recurrent clustered regimes, including reference-mode occupancy, swap frequencies, and morphology-axis diagnostics. Table A1 reproduces the cohort-level cluster list used in the main Results. Table A2 records the mapping between recurrent rank clusters and their dominant reference-mode identities. Table A3 reports per-mode swap rates, Table A4 gives the adjacent rank-pair diagnostics, Table A5 summarises the exchange-regime classification, and Table A6 reports the morphology-axis permutation null test used to localise swap behaviour.

Table A1. Detected near-degenerate rank clusters

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	0.4331	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	0.1729	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	0.1588	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

Table A2. Rank-cluster to reference-mode mapping

rank_cluster	dim	n_subjects	prevalence	paired_all_rate	ref_modes_mode	ref_modes_mode_rate	ref_modes_top3
Rank modes 9–10	2	152	54.7%	1	9–10	1	9–10 (100%)
Rank modes 12–15	4	116	41.7%	0.7414	12–15	1	12–15 (100%)
Rank modes 5–6	2	94	33.8%	1	5–6	0.8298	5–6 (83%); 4,6 (17%)
Rank modes 12–13	2	78	28.1%	0.9872	13–14	0.4805	13–14 (48%); 12–13 (44%); 12,15 (6%)
Rank modes 14–15	2	70	25.2%	0.8571	12,15	0.45	12,15 (45%); 14–15 (38%); 13–14 (15%)
Rank modes 12–14	3	45	16.2%	0.9333	12–14	0.8571	12–14 (86%); 12,13,15 (10%); 13–15 (5%)
Rank modes 4–5	2	35	12.6%	1	4–5	1	4–5 (100%)
Rank modes 13–15	3	29	10.4%	0.8966	12–14	0.5	12–14 (50%); 13–15 (31%); 12,14,15 (15%)
Rank modes 7–8	2	21	7.6%	1	7–8	1	7–8 (100%)

Table A3. Per-mode swap rates

Mode	Swap rate	Fail rate	Non-id. rate	n swapped	n failed
1	0	0	0	0	0
2	0	0	0	0	0
3	0.003597	0	0.003597	1	0
4	0.2698	0	0.2698	75	0
5	0.3777	0	0.3777	105	0
6	0.1835	0	0.1835	51	0
7	0.01079	0	0.01079	3	0
8	0.003597	0	0.003597	1	0
9	0.2698	0	0.2698	75	0
10	0.2698	0	0.2698	75	0
11	0.003597	0	0.003597	1	0
12	0.6151	0.007194	0.6223	171	2
13	0.6295	0	0.6295	175	0
14	0.5072	0.01439	0.5216	141	4
15	0.3813	0.1978	0.5791	106	55

Table A4. Adjacent rank-pair swap diagnostics

cluster	cluster_key	rank_pair	ref_pair_mode	swap_rate	invalid_rate	match_rate	median_gap_in	near_deg_rate	true_deg_rate	peak_rate	active_frac	cv_swap	verdict
Rank modes 9–10	modes_9_10	9 $\rightleftharpoons$ 10	9 $\rightleftharpoons$ 10	0.3750	0.0000	1.0000	0.1206	0.5468	0.0000	1.0000	0.2050	0.6157	veering
Rank modes 12–15	modes_12_15	12 $\rightleftharpoons$ 13	12 $\rightleftharpoons$ 13	0.4154	0.0345	0.5603	0.0516	0.8597	0.0000	1.0000	0.3094	0.7588	veering
Rank modes 12–15	modes_12_15	13 $\rightleftharpoons$ 14	13 $\rightleftharpoons$ 14	0.1429	0.0603	0.3017	0.0802	0.7014	0.0000	1.0000	0.0324	2.0521	veering
Rank modes 12–15	modes_12_15	14 $\rightleftharpoons$ 15	14 $\rightleftharpoons$ 15	0.1818	0.2241	0.3793	0.0721	0.7734	0.0000	1.0000	0.0935	1.6757	veering
Rank modes 5–6	modes_5_6	5 $\rightleftharpoons$ 6	5 $\rightleftharpoons$ 6	0.1282	0.0000	0.8298	0.1661	0.3633	0.0000	1.0000	0.0288	1.6080	veering
Rank modes 12–13	modes_12_13	12 $\rightleftharpoons$ 13	13 $\rightleftharpoons$ 14	0.0811	0.0128	0.4744	0.0516	0.8597	0.0000	1.0000	0.0432	2.9996	veering
Rank modes 14–15	modes_14_15	14 $\rightleftharpoons$ 15	12 $\rightleftharpoons$ 15	0.6296	0.1429	0.3857	0.0721	0.7734	0.0000	1.0000	0.3921	0.8933	veering

Table A5. Cluster-level exchange classification

cluster	cluster_key	dim	n_subjects	prevalence	median_gap_in_rate	gap_type	swap_verdict	final_verdict	ref_mode_mode	ref_mode_mode_rate	paired_all_rate	near_swap_rate	failed_rate	within_block_rate	transposition_rate	cycle_gd2_rate	n_dig_gains	n_covering_gains
Rank modes 9–10	modes_9_10	2	152	54.7%	0.0011	near-degenerate	VEERING (localized character exchange)	VEERING (localized character exchange)	9–10	1.0000	1.0000	0.3750	0.0000	1.0000	0.3750	0.0000	1	1
Rank modes 12–15	modes_12_15	4	116	41.7%	0.0015	near-degenerate	VEERING (localized character exchange)	MIXING (multi-mode; cycles ≤ 2 common)	12–15	1.0000	0.7414	0.5431	0.2586	0.7414	0.3023	0.3072	3	3
Rank modes 5–6	modes_5_6	2	94	33.8%	0.0074	near-degenerate	VEERING (localized character exchange)	VEERING (localized character exchange)	5–6	0.8298	1.0000	0.1064	0.0000	0.8298	0.1282	0.0000	1	1
Rank modes 12–13	modes_12_13	2	78	25.1%	0.0084	near-degenerate	VEERING (localized character exchange)	VEERING (localized character exchange)	13–14	0.4805	0.9872	0.0305	0.0128	0.4744	0.0811	0.0000	1	1
Rank modes 14–15	modes_14_15	2	70	25.2%	0.0012	near-degenerate	VEERING (localized character exchange)	VEERING (localized character exchange)	12–15	0.4500	0.8571	0.1429	0.1429	0.3857	0.3704	0.0000	1	1

Table A6. Morphology-axis permutation null test

Cluster	Rank pair	Peak _{obs}	p (peak)	Active _{obs}	p (active)	CV _{obs}	p (CV)	Significant
Rank modes 9–10	9 $\rightleftharpoons$ 10	1	0.2547	0.2162	0.8202	0.6042	0.02697	NO
Rank modes 12–15	12 $\rightleftharpoons$ 13	1	0.4496	0.283	0.7223	0.5553	0.4046	NO
Rank modes 12–15	13 $\rightleftharpoons$ 14	0.25	0.8891	0.5	0.3437	1	0.4336	NO
Rank modes 12–15	14 $\rightleftharpoons$ 15	0.5	0.7443	0.09302	0.8492	0.9727	0.4256	NO
Rank modes 5–6	5 $\rightleftharpoons$ 6	0.5	0.9001	0.1267	0.4535	1.048	0.7263	NO
Rank modes 12–13	12 $\rightleftharpoons$ 13	0.25	0.8192	0.4375	0.2438	1.146	0.5455	NO
Rank modes 14–15	14 $\rightleftharpoons$ 15	0.75	0.6563	1	0.5504	0	1	NO

B Cohort robustness and sex effects

The main text uses sex effects only where they materially sharpen interpretation of the recurrent clustered subspaces, especially the 9–10 inlet-related pair. This appendix provides the broader sex-stratified summary and the bootstrap evidence that the dominant clusters are stable cohort-level features rather than products of a few outlying subjects. Table B1 gives the bootstrap stability estimates for the dominant clusters and gap statistics. Figure B1 summarises the sex-stratified spectral structure, and Table B2 reports the accompanying sex-stratified statistical tests for the spectral and subspace metrics.

Table B1. Bootstrap robustness under random 80% subcohorts

Metric	Target	Observed	Boot. mean	Boot. std	95% CI
prevalence	Rank modes 9–10	0.547	0.548	0.015	[0.518, 0.577]
prevalence	Rank modes 12–15	0.417	0.417	0.015	[0.387, 0.450]
prevalence	Rank modes 5–6	0.338	0.338	0.014	[0.311, 0.365]
prevalence	Rank modes 12–13	0.281	0.280	0.014	[0.252, 0.311]
prevalence	Rank modes 14–15	0.252	0.252	0.013	[0.225, 0.275]
swap_rate	Mode 4	0.270	0.270	0.014	[0.243, 0.295]
swap_rate	Mode 5	0.378	0.379	0.015	[0.351, 0.405]
swap_rate	Mode 6	0.183	0.183	0.012	[0.158, 0.207]
swap_rate	Mode 7	0.011	0.011	0.003	[0.005, 0.014]
swap_rate	Mode 9	0.270	0.270	0.013	[0.243, 0.293]
swap_rate	Mode 10	0.270	0.270	0.013	[0.243, 0.293]
swap_rate	Mode 12	0.594	0.593	0.014	[0.565, 0.622]
swap_rate	Mode 13	0.622	0.622	0.015	[0.592, 0.653]
swap_rate	Mode 14	0.468	0.467	0.015	[0.439, 0.500]
swap_rate	Mode 15	0.453	0.453	0.015	[0.423, 0.484]
median_gap	Pair 1–2	1.609	1.608	0.015	[1.572, 1.634]
median_gap	Pair 2–3	1.183	1.186	0.010	[1.169, 1.206]
median_gap	Pair 3–4	0.484	0.483	0.008	[0.465, 0.495]
median_gap	Pair 4–5	0.271	0.271	0.005	[0.261, 0.281]
median_gap	Pair 5–6	0.166	0.166	0.003	[0.156, 0.170]
median_gap	Pair 6–7	0.526	0.527	0.005	[0.516, 0.539]
median_gap	Pair 7–8	0.287	0.287	0.005	[0.277, 0.297]
median_gap	Pair 8–9	0.913	0.912	0.007	[0.895, 0.922]
median_gap	Pair 9–10	0.121	0.121	0.003	[0.117, 0.127]
median_gap	Pair 10–11	1.144	1.146	0.012	[1.129, 1.165]
median_gap	Pair 11–12	0.705	0.704	0.005	[0.694, 0.712]
median_gap	Pair 12–13	0.052	0.052	0.002	[0.049, 0.056]
median_gap	Pair 13–14	0.080	0.080	0.002	[0.076, 0.085]
median_gap	Pair 14–15	0.072	0.071	0.003	[0.065, 0.076]

Figure B1. Sex-stratified spectral summary

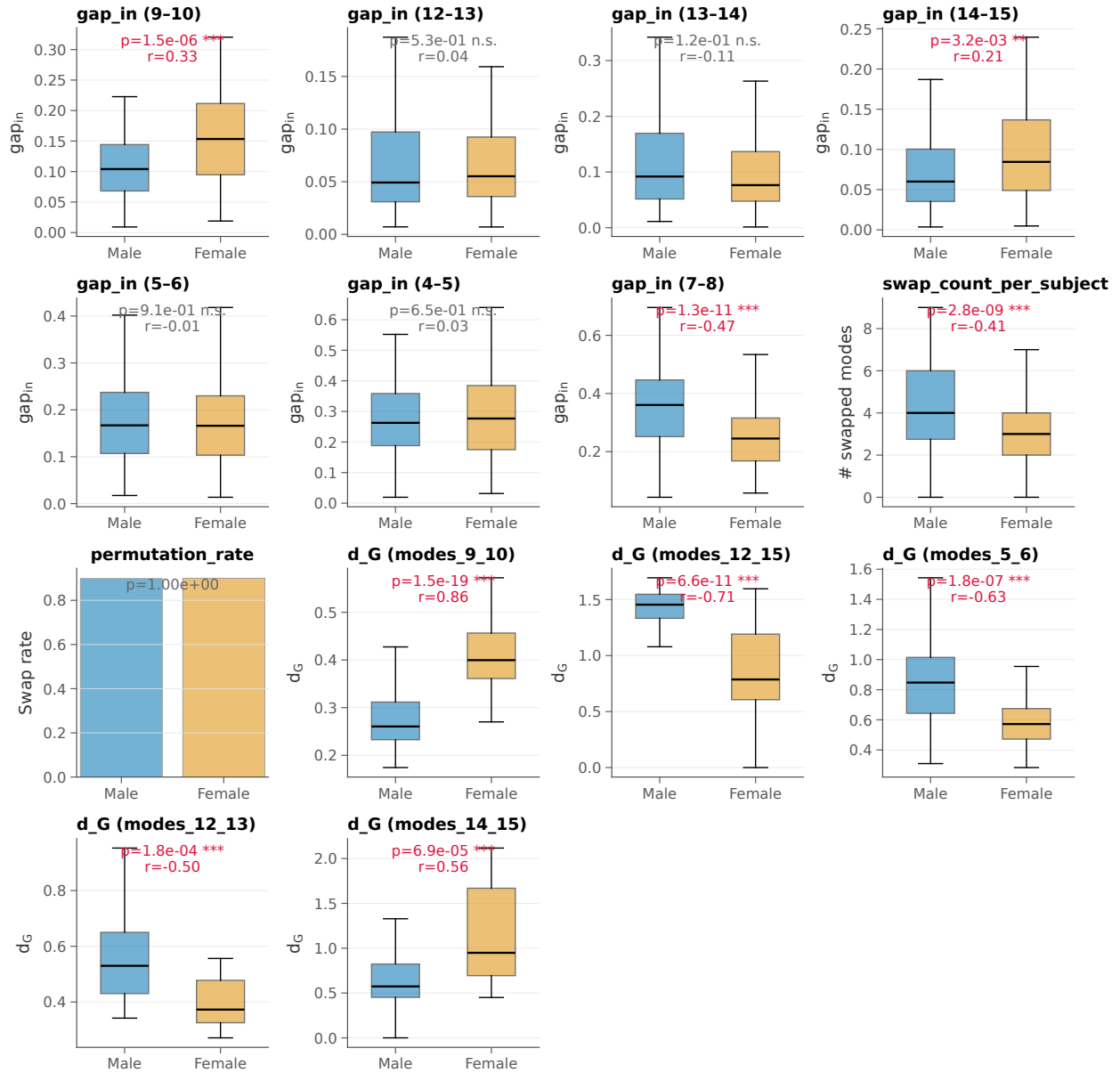


Table B2. Sex-stratified tests for spectral and subspace metrics

Metric	Male med.	Female med.	U	p	r_{rb}
gap_in (9–10)	0.104	0.1534	6384	1.489e-06	0.335
gap_in (12–13)	0.04925	0.0552	9182	0.5321	0.04354
gap_in (13–14)	0.09191	0.07643	1.063e+04	0.1233	-0.1073
gap_in (14–15)	0.06002	0.08448	7631	0.003216	0.2051
gap_in (5–6)	0.1669	0.166	9677	0.9088	-0.008021
gap_in (4–5)	0.2627	0.2768	9299	0.6529	0.03135
gap_in (7–8)	0.3604	0.245	1.412e+04	1.311e-11	-0.471
swap_count_per_subject	4	3	1.352e+04	2.824e-09	-0.4079
permutation_rate	0.8984	0.9	0.9829	1	-0.01724
d_G (modes_9_10)	0.2605	0.3996	377	1.5e-19	0.8649
d_G (modes_12_15)	1.455	0.7866	2848	6.629e-11	-0.7059
d_G (modes_5_6)	0.8472	0.572	1793	1.766e-07	-0.6263
d_G (modes_12_13)	0.5299	0.3729	1120	0.0001815	-0.4973
d_G (modes_14_15)	0.5736	0.9485	270	6.922e-05	0.5559

C Load-routing specificity

The main manuscript uses standing and proxy childbirth loads to show the contrast between a robust low-rank backbone and selectively recruited higher-rank subspaces. This appendix provides the fuller load-specificity diagnostics used during editorial triage, including cluster-level function-locking, dominant-mode sets, and pairwise overlap of full routing profiles. Figure C1 visualises the load–function locking map across the prevalent clusters. Table C1 summarises the phase-specific coupling synthesis, Table C2 reproduces the dominant mode-set summary used in the main text for appendix-level completeness, and Table C3 reports the pairwise overlap metrics for the full load-routing profiles.

Figure C1. Load-function coupling heatmap across prevalent clusters

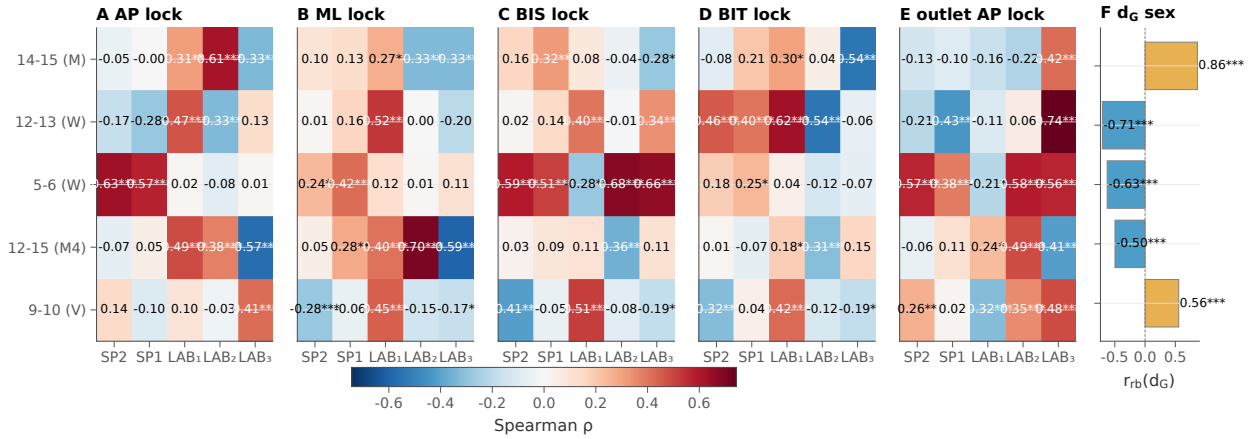


Table C1. Load-function coupling synthesis

Cluster	Regime	Best-predictor coupling axis for FLI (AP/ML inlet, BIS/BIT outlet, or Outlet AP scalar)	FLI ρ	Load case	$\bar{f}_C$	$r_b(d_G)$
Rank modes 9–10	Veering	BIS outlet	0.5144	LAB ₁	0.5291	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.3508	LAB ₂	0.1578	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.4828	LAB ₃	0.1407	0.8649
Rank modes 12–15	Mixing (4D)	AP inlet	0.4858	LAB ₁	0.07568	-0.7059
Rank modes 12–15	Mixing (4D)	ML inlet	0.7	LAB ₂	0.01016	-0.7059
Rank modes 12–15	Mixing (4D)	ML inlet	-0.5942	LAB ₃	0.06857	-0.7059
Rank modes 5–6	Weak exchange	BIS outlet	-0.2788	LAB ₁	0.02502	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.6786	LAB ₂	0.5917	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.6589	LAB ₃	0.217	-0.6263
Rank modes 12–13	Weak exchange	BIT outlet	0.6247	LAB ₁	0.06863	-0.4973
Rank modes 12–13	Weak exchange	BIT outlet	-0.5387	LAB ₂	0.0009939	-0.4973
Rank modes 12–13	Weak exchange	Outlet AP scalar	0.7433	LAB ₃	0.01162	-0.4973
Rank modes 14–15	Mixing	AP inlet	0.3085	LAB ₁	0.02817	0.5559
Rank modes 14–15	Mixing	AP inlet	0.6098	LAB ₂	0.01237	0.5559
Rank modes 14–15	Mixing	BIT outlet	-0.54	LAB ₃	0.008386	0.5559

Table C2. Dominant mode sets for standing and proxy loading

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB ₁	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB ₂	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB ₃	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

Table C3. Pairwise overlap of load-routing profiles

Load A	Load B	Jaccard overlap (80% sets)	Cosine similarity (full profile)	Shared modes (80% sets)
SP2leg	SP1leg	0.3333	0.6668	2
SP2leg	LAB ₁	0	0.1202	
SP2leg	LAB ₂	0	0.1343	
SP2leg	LAB ₃	0.2	0.9367	2
SP1leg	LAB ₁	0.1429	0.1233	3
SP1leg	LAB ₂	0	0.1051	
SP1leg	LAB ₃	0.1429	0.6326	2
LAB ₁	LAB ₂	0.2857	0.48	4,9
LAB ₁	LAB ₃	0.25	0.3613	4,9
LAB ₂	LAB ₃	0.8	0.4456	4,5,6,9

D Model-boundary and sensitivity analyses

These analyses bound the main claims rather than extend them. Allometry quantifies what is lost when the main spectral analysis is performed on size-normalised pelves. Material uncertainty quantifies how much absolute eigenvalue identifiability would degrade under broader soft-tissue and density-law scatter while showing that the relative fragile-versus-robust gap architecture is preserved. Figure D1 shows the post-hoc allometric recovery analysis, and Table D1 summarises the corresponding scaling exponents and variance decomposition. Figure D2 presents the threshold sensitivity, reference sensitivity, isotropic scale vs solver gap, and 15-mode displacement reconstruction error, alongside Table D2 which tabulates the combined first-order uncertainty propagation used to support the robustness discussion in the main text.

Figure D1. Allometry dashboard

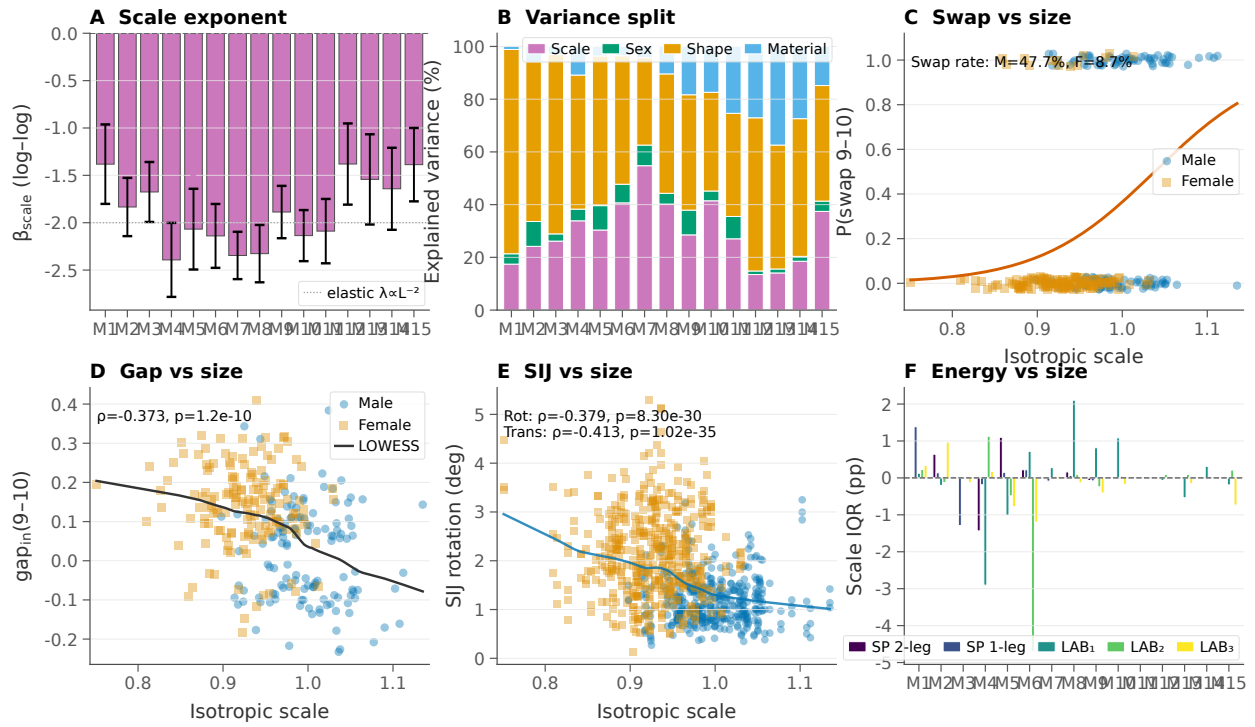


Table D1. Allometry summary

Table 3: Allometric scaling exponents and variance decomposition per eigenvalue mode. β_{scale} is the IV-2SLS log-log coefficient of eigenvalue on isotropic scale, with HC-robust 95 % CIs and FDR-corrected p -values. $\Delta\lambda$ per IQR gives the percent eigenvalue change for an interquartile range shift in log-scale. R^2 columns show the LMG (Shapley) variance share for each predictor group.

Mode	β_{scale}	95% CI	p_{FDR}	$\Delta\lambda$ IQR (%)	R^2 scale (%)	R^2 shape (%)	R^2 material (%)	R^2 sex (%)
1	-1.38	[-1.80, -0.96]	4.7e-10	-10.1	17.4	77.5	1.1	3.9
2	-1.83	[-2.14, -1.53]	8.4e-26	-13.1	24.2	60.6	5.8	9.4
3	-1.68	[-1.99, -1.36]	2.0e-21	-12.1	26.2	68.0	3.1	2.7
4	-2.39	[-2.78, -2.00]	7.7e-27	-16.8	33.9	50.8	10.9	4.4
5	-2.07	[-2.49, -1.64]	1.1e-18	-14.7	30.4	56.8	3.6	9.3
6	-2.14	[-2.48, -1.80]	3.4e-28	-15.1	40.7	48.8	3.5	7.1
7	-2.35	[-2.60, -2.10]	5.7e-49	-16.5	54.8	33.1	4.3	7.8
8	-2.33	[-2.63, -2.02]	2.4e-37	-16.4	40.2	45.3	10.4	4.1
9	-1.89	[-2.16, -1.61]	1.9e-31	-13.5	28.5	43.8	18.3	9.4
10	-2.14	[-2.41, -1.87]	7.7e-39	-15.1	41.5	37.4	17.4	3.7
11	-2.09	[-2.43, -1.75]	6.3e-27	-14.8	27.0	39.1	25.4	8.5
12	-1.38	[-1.81, -0.95]	9.4e-10	-10.1	13.5	58.1	27.1	1.3
13	-1.54	[-2.02, -1.07]	8.2e-10	-11.2	14.1	47.0	37.4	1.5
14	-1.64	[-2.07, -1.21]	1.5e-12	-11.8	18.5	52.2	27.4	1.9
15	-1.39	[-1.77, -1.00]	1.8e-11	-10.1	37.4	43.8	14.8	3.9

Figure D2. Robustness and reconstruction dashboard

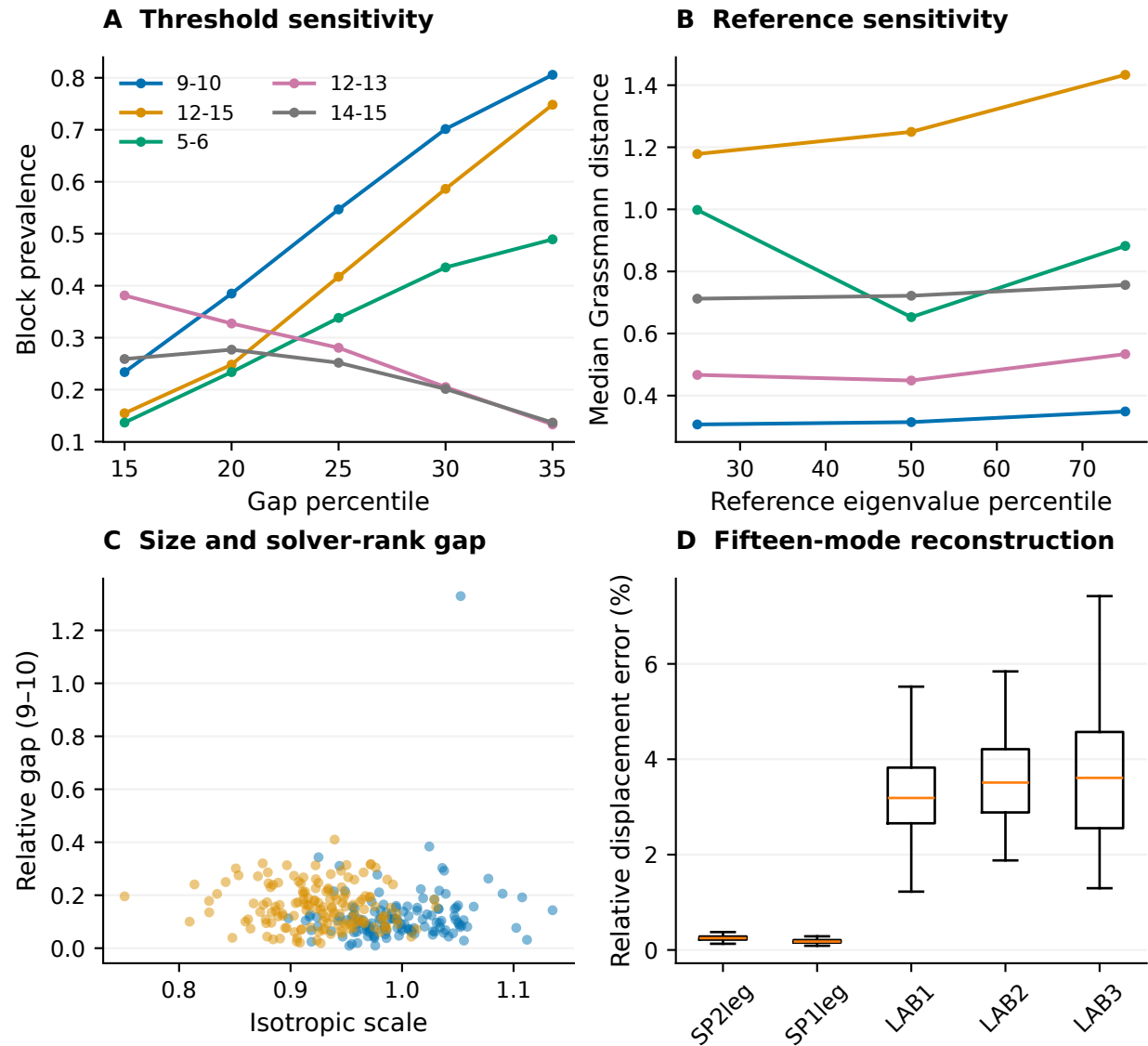


Table D2. Combined first-order material uncertainty summary

Table 4: Population-level material uncertainty propagation stratified by sex. CoV is the total coefficient of variation (ligament + cartilage + bone). Lig/Cart/Bone columns show each group’s fraction of total variance. Gap Z -score measures robustness of each spectral gap to combined material noise (first-order signal-to-noise ratio; not a confidence probability).

Mode	Female ($N = 150$)				Male ($N = 128$)			
	CoV (%)	Lig (%)	Cart (%)	Bone (%)	CoV (%)	Lig (%)	Cart (%)	Bone (%)
1	46.0	5	94	0	46.5	7	93	1
2	46.5	4	93	2	46.7	5	93	2
3	42.5	5	94	1	43.3	6	93	1
4	46.0	6	90	4	45.1	8	90	2
5	44.2	8	89	3	44.2	9	89	2
6	44.9	9	89	2	44.7	10	88	2
7	45.9	16	83	1	45.5	15	84	1
8	47.3	7	90	3	46.9	6	91	3
9	45.3	4	88	7	43.9	4	90	5
10	43.8	2	90	8	44.1	5	90	6
11	44.5	1	87	12	43.7	3	89	9
12	45.7	1	85	14	45.4	2	85	13
13	46.3	1	84	15	46.1	1	83	16
14	45.4	1	86	13	45.2	2	87	10
15	45.4	2	88	10	45.0	3	89	8

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